

Yeh-Ching Chung, Professor
School of Data Science
Chinese University of Hong Kong in Shenzhen
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Research Interests

- Cloud-Fog Computing: Virtualization Techniques (CPUs, GPUs, Memory, Storage, and Network), Cloud-RAN Design for 5G/IoT Networks
 - Big Data: Applications (Agriculture and Large-Scale Software System Optimizations), Graph Computing for Association Rule Mining, and Deep learning for Medical Image Processing
 - Embedded Systems: Tool-Chain for Heterogeneous System Architecture (Compilers, Emulators, and Debuggers), Virtualization for Heterogeneous System Architecture, Embedded Virtual Platform
 - Parallel and Distributed Systems: Parallelizing Compilers, Parallel I/O and File Systems, P2P, HLA, and Grid Computing
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Education

- Ph.D. in Computer and Information Science (1992), Syracuse University
 - M.S. in Computer Science (1988), Syracuse University
 - B.S. in Computer Science (1983), Chung-Yuan Christian University
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Experience

- Professor – School of Data Science, Chinese University of Hong Kong in Shenzhen, Shenzhen, Guangdong 518172, China (Since July 2020)
- Professor – School of Science and Engineering, Chinese University of Hong Kong in Shenzhen, Shenzhen, Guangdong 518172, China (September 2018 – June 2020)
- Deputy Director – Cloud Computing and Disaster Recovery Laboratory, Research Institute of Tsinghua University in Shenzhen, Shenzhen, Guangdong 518057, China (October 2015 – August 2018)
- Professor – Department of Computer Science, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (August 2002 – January 2016)
- Deputy Director – Library, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (December 2003 – July 2012)
- Deputy Director – Computer and Communication Research Center, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (August 2008 – July 2015)
- Director General – Taiwan Association of Cloud Computing (March 2007 – February 2011)

- Chairman – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (August 1998 – July 2001)
- Professor – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (July 1999 – July 2002)
- Associate Professor – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (August 1992 – June 1999)
- Director – Networking Division, Computer Center, Feng Chia University, Taichung, Taiwan 407, ROC (January 1994 – July 1997)
- Director – Education Division, Computer Center, Feng Chia University, Taichung, Taiwan 407, ROC (August 1993 – December 1993)

Professional Societies Membership

- IEEE Senior Member, IEEE Computer Society
- ACM, ACM SIGBED
- Taiwan Association Cloud Computing (TACC)

Awards

- 2024 ACM SCC conference - The Outstanding Paper Award. Paper title “Isolate and Detect the Untrusted Driver with a Virtual Box”
- 2024 Outstanding Alibaba Innovative Research Program – Project title “A Large-Scale Graph Pattern Matching Query System and Optimizations”
- 2021 CUHK-SZ Ethics of Teachers Construction Award
- 2017 IEEE SC2 conference - The Outstanding Paper Award. Paper title “Application-Aware Traffic Redirection: A Mobile Edge Computing Implementation Toward Future 5G Networks”
- 2017 IEEE ICASI conference - The First Prize Paper Award. Paper title “File Placement Mechanisms for Improving Write Throughputs of Cloud Storage Services Based on Ceph and HDFS”
- 2016 IEEE SC2 conference - The Best Poster Award. Poster title "Byzantine Fault Tolerant Optimization in Federated Cloud Computing"
- 2016 ACM VEE conference - The Outstanding Paper Award. Paper title “Building A KVM-based Hypervisor for A Heterogeneous System Architecture Compliant System”
- 2015 China Thousand Talents Plan Awardee (innovative long-term)
- 2014 NTHU Distinguished Industry-Academy Cooperation Award for excellent contribution to the Big Data
- 2014 ACM/IEEE Supercomputing conference (SC'14) - Student Cluster Competition Highest LINKPACK Award

- 2014 NCHC Student Cluster Competition Champion
- 2013 NCHC Student Cluster Competition Champion
- 2012 NTHU Distinguished Industry-Academy Cooperation Award for excellent contribution to the Cloud Computing
- 2012 NCHC Student Cluster Competition Champion
- 2011 IEEE ICPADS conference - The Best Paper Award. Paper title "PQEMU: A Parallel System Emulator Based on QEMU"
- 2011 ACM/IEEE Supercomputing conference (SC'11) - Student Cluster Competition Overall Winner (Champion)
- 2011 NCHC Student Cluster Competition Champion
- 2010 ACM/IEEE Supercomputing Conference (SC'10) - Student Cluster Competition overall winner (Champion)
- 2010 Embedded System – Hardware/Software Co-Design Contest Champion, Ministry of Education of ROC
- 2008 Chung-Shang Institute of Science and Technology Distinguished Academy Cooperation Award for excellent contribution to the High Level Architecture (HLA) System Design

Grants (Since 2003)

- The Key Techniques of Workload Generator on Cloud Environments, Huawei , RMB1,190,000, 2022.11-2024.04.
- A Large-Scale Graph Matching Query System and Optimizations, Ali Damou Research Institute, RMB 500,000, 2022.05-2023.04.
- An Agile and Customizable Intelligent Visual Processor and its Applications, Key Area Research and Development Program of Guangdong Province, No. 2019B010155002, RMB 20,000,000, 2020.01-2024.01. (with C.W. Chen et al.)
- A Generic Graph Computing Processor Technique and System, Ministry of Science and Technology, RMB 40,610,000, 2018.05-2020.05. (with X.F. Liao et al.)
- A High Reliable Communication Theory and Technology Under Ocean Environment, NSFC Guangdong joint fund, RMB 2,700,000, 2018.01-2022.01. (with K.Q. Li et al.)
- A Fusion of SQL and NoSQL Databases for Big Data Applications, Guangdong Province, RMB 1,000,000, 2016.10-2018.10.
- A High-Performance Big Data Analytical Platform Based on Graph Computing, Shenzhen City Brach Committee, RMB 1,000,000, 2016.01-2018.01. (with W.G. Chen)
- Smart City, Ministry of Science and Technology, NT\$5,000,000, 2014.08-2017.08. (with C.T. King et al.)
- 5G/ToT Cloud Platform, ITRI, NT\$2,000,000, 2014.10-2016.04.
- High-Performance Big Data Appliance, National Science Council, NT\$8,000,000, 2014.02-2015.02.

- A Hybrid Just-in-Time Compilation for Android System, National Science Council, NT\$800,000, 2013.08-2014.08.
- The Automatic Workflow Deployment on Cloud Platform, III, NT\$800,000, 2013.04-2014.04.
- Virtualization Techniques for ARM Processors, National Science Council, NT\$26,000,000, 2012.12-2016.12.
- Cloud Platforms for HPC Applications, Quanta Computer Corporation, NT\$7,900,000, 2012.08-2013.07
- Distributed File System, National Science Council, NT\$761,000, 2012.08-2013.07
- DCloud - A Distributed Cloud Middleware, Delta Electronic Corporation, NT\$36,000,000, 2011.08-2014.07.
- Development of HLA-based Simulation System using Distributed Computing Techniques, National Science Council, NT\$797,000, 2011.01-2011.12.
- UniCloud: A Distributed Cloud Computing Middleware, National Science Council, NT\$1,349,000, 2010.08-2011.07.
- UniCloud-Subproject C1: Storage Virtualization, National Science Council, NT\$945,000, 2010.08-2011.07.
- A Volunteer Computing System for Computer Games and its Applications, National Science Council, NT\$2,528,000, 2010.08-2011.07. (with I.C. Wu et al.)
- The Design of Desktop Grids in the Volunteer Computing System, National Science Council, NT\$2,206,000, 2010.08-2013.07. (with I.C. Wu et al.)
- Ontology on Monte Carlo Tree Search and Applications, National Science Council, NT\$7,181,000, 2010.01-2012.12. (with C.S. Lee et al.)
- A P2P HLA RTI Design, National Science Council, NT\$851,000, 2009.01-2009.12.
- Home-Grid: A Pervasive P2P Based Personal Health-Care Grid System, National Science Council, NT\$20,171,000, 2008.11-2011.10.
- NTHU-MTK Embedded System Lab, MediaTek Corporation, NT\$60,000,000, 2008.08-2013.07. (with J.M. Shyu et al.)
- ESL Based Developing and Verification Environment for Embedded Multicore Systems, National Science Council, NT\$3,226,000, 2008.08-2010.07. (with J.M. Shyu et al.)
- Pisces: A P2P Meta-Grid System, National Science Council, NT\$3,260,000, 2007.08-2010.07.
- Virtual Battlefield, Chung-Shang Institute of Science and Technology, NT\$4,342,000, 2007.01-2008.12.
- A Wireless Sensor Network for Landslides Monitoring, National Science Council, NT\$12,884,000, 2006.11-2009.10. (with Pai Chou et al.)
- MultiFed – A Scalable HLA RTI System, Chung-Shang Institute of Science and Technology, NT\$4,220,000, 2006.01-2008.12.
- Medicare Grid – A Grid Based E-health System, National Science Council, NT\$20,802,000, 2005.11-2008.10.
- Design a Fat-Tree Based Cluster System, National Science Council, NT\$1,221,000, 2005.08-2007.07.

- An Intelligent Sensor Network and its Applications, Ministry of Economic Affairs, NT\$69,000,000, 2005.06-2008.05. (with C.T. King et al.)
- StarFish – An Ultralow power DSP, Ministry of Economic Affairs, NT\$55,000,000, 2005.06-2008.05. (with Y.L. Lin et al.)
- Taiwan UniGrid – A University Grid In Taiwan, National Center for High-Performance Computing), NT\$4,040,000, 2005.01-2006.12.
- A General-Purpose Sensor Platform Design, National Science Council, NT\$4,821,100, 2004.08-2007.07.
- Overlay Network Technology and Applications, National Science Council, NT\$27,042,400, 2004.04-2008.03. (with C.T. King et al.)
- A Data Grid System for Data Streaming Applications Based on Microsoft Windows, Microsoft, NT\$1,800,000, 2004.01-2005.01
- HPC Technology and Applications Based on Microsoft Windows, Microsoft, NT\$2,400,000, 2003.01-2004.01.

Journal Editorship

- Associate Editor, ACM Computing Surveys, Since 2014
- Associate Editor, International Journal of Grid and High-Performance Computing (IJGHPC), Since 2009
- Associate Editor, International Journal of Big Data Intelligence (IJBDI), 2013-2015
- Advisory Committee Member, DataCom conference, since 2019
- Associate Editor, Journal of Chinese Institute of Engineers (JCIE), 2012-2013

Patents

- **Yeh-Ching Chung** and Yuan Yu, “Terminal, Storage Medium, and Database Synchronization Method Thereof,” China Patent, No. ZL 2020 1 0933547.2, November 07, 2023.
- **Yeh-Ching Chung** and Yuan Yu, “Terminal, Storage Medium, and Database Synchronization Method Thereof,” US Patent, No. US11604805 B2, March 14, 2023.
- **Yeh-Ching Chung**, Lidong Zhang, and Yongwei Wu, “Hybrid File System Architecture, File Storage, Dynamic Migration, and Application Thereof,” US Patent, No. US10810169 B1, Oct.20, 2020.
- **Yeh-Ching Chung**, Rui Zhang, and Wenguang Chen, “Method for Computing Support of Itemset Candidate Based on Graph Structure Data and Application Thereof,” US Patent, No. US10776372 B2, Sept. 15, 2020.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, “Synchronization System for Transforming Database and Method Thereof,” China Patent, No. 201410776112, October 25, 2018.

- Hung-Chang Hsiao, Chi-Tsun Liao, Chia-Ping Tsai, and **Yeh-Ching Chung**, “System and method for recovering system state consistency to any point-in-time in distributed database,” Taiwan patent. No. I626547, Taiwan, June 11, 2018.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, “Synchronization System for Transforming Database and Method Thereof,” US patent, No. US9779154B2, October 3, 2017.
- Hung-Chang Hsiao, Chi-Tsun Liao, Chia-Ping Tsai, and **Yeh-Ching Chung**, “System and method for recovering system state consistency to any point-in-time in distributed database,” US patent, No. US9372725B2, June 21, 2016.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, “Synchronization System for Transforming Database and Method Thereof,” Taiwan patent, No. I539309, June 21, 2016.
- **Yeh-Ching Chung** and Jiun-Hung Ding, “Multicore Processing System and its Scheduling Method,” Taiwan patent, No. I503742, October 11, 2015.
- Wei-Chih Sun and **Yeh-Ching Chung**, “Cloud Management Systems and Methods for Executing Applications on Android Systems,” Taiwan patent, No. I49918, September 9, 2015.

Book Chapters

1. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, “Effective Resource Allocation and Job Scheduling Mechanisms for Load Sharing in a Computational Grid”, *Handbook of Research on Grid Technologies and Utility Computing: Concepts for Managing Large-Scale Applications*, Edited by Emmanuel Udoh and Frank Zhigang Wang, pp. 31-40, Information Science Publishing, May 31, 2009, (ISBN: 978-1605661841).
 2. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, "QoS-based Job Scheduling and Resource Management Strategies for Grid Computing," *Quantitative Quality of Service for Grid Computing: Applications for Heterogeneity, Large-Scale Distribution, and Dynamic Environments*, Edited by Lizhe Wang, Jinjun Chen, and Wei Jie, pp. 110-127, Information Science Publishing, May 5, 2009 (ISBN: 978-1605663708).
 3. William C. Chu, Chih-Hong Chang, Chih-Wei Lu, H. Yang, H.C. Jiau, **Yeh-Ching Chung**, and B. Qiao, “Enhanceing Software Maintainability by Unifying and Integrating Standards,” *Advances in Software Maintenance Management: Technologies and Solutions*, pp. 114-150, Idea Group Publishing, 2002.
 4. Chih-Wei Lu, William C. Chu, Chih-Hong Chang, **Yeh-Ching Chung**, X. Liu, and H. Yang, “Reverse Engineering,” *Handbook of Software Engineering and Knowledge Engineering*, Vol. 2, pp. 447-466, 2002.
 5. William C. Chu, Chih-Wei Lu, Chih-Hong Chang, and **Yeh-Ching Chung**, “Pattern-Based Software Reengineering,” *Handbook of Software Engineering and Knowledge Engineering*, Vol. 1, pp.767-786, 2001.
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Journal Papers

- J1. Lele Li, Wei-Chung Hsu, **Yeh-Ching Chung**, and Zhibin Yu, "Improving Resource Efficiency of Performance-optimized In-memory Big Data Analytics with a Hybrid Static Dynamic Approach," *IEEE Transactions on Computers* (TC), Vol. 75, No. 9, September 2026, pp. 3271-3283. (SCI, EI, CCF A)
- J2. Shihao Hong and **Yeh-Ching Chung**, "CRPIM: An Efficient Compute-Reuse Scheme for ReRAM-based Processing-in-Memory DNN Accelerators," *Journal of System Architecture* (JSA), 153, August 2024, 103192. (SCI, EI, CCF B)
- J3. YongGang Li, Yu Bao, and **Yeh-Ching Chung**, "Randomize the Running Function When It Is Disclosed," *IEEE Transactions on Computers* (TC), Vol. 73, No. 6, 2024, pp. 1516-1530. (SCI, EI, CCF A)
- J4. YongGang Li, Jiazhen Cai, Yu Bao, and **Yeh-Ching Chung**, "What You Can Read is What You Can't Execute," *Computers and Security* (COSE), Vol. 132, July 2023, 103377. (SCI, EI, CCF B)
- J5. YongGang Li, Jiazhen Cai, Yu Bao, and **Yeh-Ching Chung**, "What You Can Read is What You Can't Execute," *Computers and Security* (COSE), Vol. 132, July 2023, 103377 (SCI, EI, CCF B)
- J6. YuAng Chen and **Yeh-Ching Chung**, "An Unequal Caching Strategy for Shared-Memory Graph Analytics," *IEEE Transaction on Parallel and Distributed Systems* (TPDS), Vol. 34, No. 3, 2023, pp. 955-967. (SCI, EI, CCF A)
- J7. YongGang Li, GuoYuan Lin, **Yeh-Ching Chung**, YaoWen Ma, Yi Lu, and, Yu Bao, "MagBox: Keep the Risk Functions Running Safely in a Magic Box," *Future Generation Computer Systems* (FGCS), 140, 2023, pp. 282-298. (SCI, EI, CCF C)
- J8. Le-Le Li, Jiang-Yi Liu, Jian-Ping Fan, Xue-Hai Qia. Kai Hwang, **Yeh-Ching Chung**, and Zhi-Bin Yu, "SOCA-DOM: A Mobile SoC Array System for Analyzing Big Data on the Move," *Journal of Computer Science and Technology* (JCST), 37(6): Nov. 2022, pp.1271-1289. (SCI, EI, CCF B)
- J9. YuAng Chen and **Yeh-Ching Chung**, "Workload Balancing via Graph Reordering on Multicore Systems," *IEEE Transaction on Parallel and Distributed Systems* (TPDS), Vol. 33, No. 5, 2022, pp. 1231-1245. (SCI, EI, CCF A)
- J10. YongGang Li, **Yeh-Ching Chung**, Yu Bao, Yi Lu, ShanQing Guo, and GuoYuan Lin, "KPointer: Keep The Code Pointers On The Stack Point To The Right Code," *Computers and Security* (COSE), Vol. 120, September 2022, 102781. (SCI, EI, CCF B)
- J11. Yonggang Li, **Yeh-Ching Chung**, Kai Hwang, and YueJin Li, "Virtual Wall: Filtering Rootkit Attacks To Protect Linux Kernel Functions," *IEEE Transactions on Computers* (TC), Vol. 70, No. 10, 2021, pp. 1640-1653. (SCI, EI, CCF A)
- J12. Ching-Hsien Hsu, Kusuma Amilinani, Hao Wu, Christophe Cerin, and **Yeh-Ching Chung**, "A Novel Scalable Network Architecture for the Evolution of Robot Swarm Networks with Dual Connectivity Application in Control-Data Plane," *Swarm and Evolutionary Computation*, Vol. 64, 100889, July 2021. (SCI, EI)
- J13. Ching-Hsien Hsu, Xing Chen, Weiwei Lin, Chuntao Jiang, Youhong Zhang, Zhifeng Hao, and **Yeh-Ching Chung**, "Effective Multiple Cancer Disease Diagnosis Frameworks for Improved Healthcare Using Machine Learning," *Measurement*, Vol. 175, 109145, April 2021. (SCI, EI)
- J14. Deverajan Ganesh Gopa, V. Muthukumaran, Ching-Hsien Hsu, Karuppiiah Marimuthu, **Yeh-Ching Chung**, and Ying-Huei Chen, "Public Key Encryption with Equality Test

- for Industrial Internet of Things System in Cloud Computing," *Transactions on Emerging Telecommunications Technologies*, DOI: 10.1002/ett.4202. (SCI, EI)
- J15. Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung**, and Ching-Hsien Hsu, "A Creative IoT Agriculture Platform for Cloud Fog Computing," *Sustainable Computing: Informatics and Systems*, Vol. 28, 100285, 2020. (SCI, EI)
- J16. Rui Zhang, Wenguang Chen, Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung**, "ANG – A Combination of Apriori and Graph Computing Techniques for Frequent Itemsets Mining," *The Journal of Supercomputing*, Vol. 75, No. 2, 2019, pp. 646-661 (SCI, EI)
- J17. Moise Convolbo, Jerry Chu, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "GEODIS: Towards the Optimization of Data Locality-Aware Jobs Scheduling in Geo-Distributed Data Centers," *Journal of Computing*, Vol. 100, No.1, January 2018, pp. 21-46. (SCI, EI). (SCI, EI)
- J18. Fang-Ping and **Yeh-Ching Chung**, "Multi-layer Ontology Based Information Fusion for Situation Awareness," *Applied Intelligence*, Vol. 46, 2017, pp. 285-307. (SCI, EI)
- J19. Ying-Ti Liao, Jiazheng Zhou, Chia-Hung Lu, Shih-Chang Chen, Ching-Hsien Hsu, Wenguang Chen, Fong Jiang, and **Yeh-Ching Chung**, "Data Adapter for Uniform Access and Transformation in NoSQL Systems," *Future Generation Computer Systems*, Vol. 65, December 2016, pp. 111-121. (SCI, EI)
- J20. Fang-Ping Pai, I-Ching Hsu, and **Yeh-Ching Chung**, "Semantic Web Technology for Agent Interoperability: A Proposed Infrastructure," *Applied Intelligence*, Vol. 44, No. 1, January 2016, pp. 1-16. (SCI, EI)
- J21. Che-Lun Hung, Yu-Shiang Lin, Chun-Yuan Lin, **Yeh-Ching Chung**, and Yi-Fang Chung, "CUDA ClustalW: An Efficient Parallel Algorithm for Progressive Multiple Sequence Alignment on Multi-GPUs," *Computational Biology and Chemistry*, Vol. 58, October 2015, pp. 62-68. (SCI, EI)
- J22. Yu-Shiang Lin, Chun-Yuan Lin, Che-Lun Hung, **Yeh-Ching Chung**, and Kual-Zheng Lee, "GPU-UPGMA: High-Performance Computing for UPGMA Algorithm Based on Graphics Processing Units," *Concurrency and Computation: Practice and Experience*, Vol. 27, No. 13, September 2015, pp. 3403-3414. (SCI, EI)
- J23. Ching-Hsien Hsu, Kenn Slagter, and **Yeh-Ching Chung**, "Locality and Loading Aware Virtual Machine Mapping Techniques for Optimizing Communications in MapReduce Applications," *Future Generation Computer Systems*, Vol. 53, June 2015, pp. 43-54. (SCI, EI)
- J24. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "An Adaptive Sampling Mechanism for Partitioning in MapReduce," *International Journal of Parallel Programming*, Vol. 43, No. 3, June, 2015, pp. 489-507. (SCI, EI)
- J25. Ssu-Hsuan Lu, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "A Scalable P2P Overlay Based on Arrangement Graph with Minimized Overhead," *Peer-to-Peer Networking and Applications*, Vol. 7, No. 4, December 2014, pp. 497-510. (SCI, EI)
- J26. Wu-Chun Chung, Hung-Pin Lin, Shih-Chang Chen, Mon-Fong Jang, and **Yeh-Ching Chung**, "JackHare: A Framework for SQL to NoSQL Translation Using MapReduce," *Automated Software Engineering*, Vol. 21, No. 4, December 2014, pp. 489-508. (SCI, EI)

- J27. Ching-Jung Hsu, Wu-Chun Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "On Maintenance of Cooperative Overlays in Multi-Overlay Networks," *IET Communications*, Vol. 8, No. 15, October 2014, pp. 2676-2683. (SCI, EI)
- J28. Shih-Hsiang Lo, Che-Rung Lee, Quey-Liang Kao, I-Hsin Chung, and **Yeh-Ching Chung**, "Improving GPU Memory Performance with Artificial Barrier Synchronization," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 25, No. 9, September 2014, pp. 2342-2352. (SCI, EI)
- J29. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "SmartJoin: A Network-Aware Multiway Join for MapReduce," *Cluster Computing*, Vol. 17, No. 3, September 2014, pp. 629-641. (SCI, EI)
- J30. Xuan-Yi Lin and **Yeh-Ching Chung**, "Master-Worker Model for MapReduce Paradigm on the TILE64 Many-Core Platform," *Future Generation Computer Systems*, Vol. 36, July 2014, pp. 19-30. (SCI, EI)
- J31. Jiun-Hung Ding, Roger Chien, Shih-Hao Hung, Yi-Lan Lin, Che-Yang Kuo, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "A Framework of Cloud-based Virtual Phones for Secure Intelligent Information Management," *International Journal of Information Management*, Vol. 34, June 2014, pp. 329-335. (SCI, EI)
- J32. Ssu-Hsuan Lu, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Effectiveness of a Replica Mechanism to Improve Availability with Arrangement Graph-Based Overlay," *Journal of Network and Computer Applications*, Vol. 41, May 2014, pp. 441-450. (SCI, EI)
- J33. Jiun-Hung Ding, Ya-Ting Chang, Zhoudong Kuo, Kuan-Ching Li, and **Yeh-Ching Chung**, "An Efficient and Comprehensive Scheduler on Asymmetric Multicore Architecture Systems," *The Journal of System Architecture*, Vol. 60, No. 3, March 2014, pp. 305-314. (SCI, EI)
- J34. Ding-Yong Hong, Jan-Jan Wu, Pen-Chung Yew, Wei-Chung Hsu, Chun-Chen Hsu, Pangfeng Liu, Chien-Min Wang, and **Yeh-Ching Chung**, "Efficient and Retargetable Dynamic Binary Translation on Multicores," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 25, No. 3, March 2014, pp. 622-632. (SCI, EI)
- J35. Ching-Hsien Hsu, Kenn Slagter, Shih-Chang Chen, **Yeh-Ching Chung**, "Optimizing Energy Consumption with Task Consolidation in Clouds," *Information Sciences*, Vol. 258, February 2014, pp. 452-462. (SCI, EI)
- J36. Po-Chi Shih, Kuo-Chan Huang, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung**, "TLA: Temporal Look-Ahead Processor Allocation Method for Heterogeneous Multi-Cluster Systems," *Journal of Parallel and Distributed Computing*, Vol. 73, No. 12, December 2013, pp. 1661-1672. (SCI, EI)
- J37. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, Daqiang Zhang, "An Improved Partitioning Mechanism for Optimizing Massive Data Analysis Using MapReduce," *The Journal of Supercomputing*, Vol. 66, No. 1, October 2013, pp. 539-555. (SCI, EI)
- J38. Wu-Chung Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Direction-Aware Resource Discovery in Large-Scale Distributed Computing Environments," *The Journal of Supercomputing*, Vol. 66, No. 1, October 2013, pp. 229-248. (SCI, EI)
- J39. Yu-Wei Chan, Feng-Tsun Chien, Ronald Y. Chang, Min-Kuan Chang, **Yeh-Ching Chung**, "Spectrum Sharing in Multi-Channel Cooperative Cognitive Radio Networks:

- A Coalitional Game Approach," *ACM/Springer Wireless Networks*, Vol. 19, No. 7, October 2013, pp. 1553-1562. (SCI, EI)
- J40. Xuan-Yi Lin, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Efficient Programming Paradigm for Video Streaming Processing on TILE64 Platform," *The Journal of Supercomputing*, Vol. 65, No. 2, August 2013, pp. 823-847. (SCI, EI)
- J41. Shih-Hsiang Lo, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung**, "Optimizing Pairwise Box Intersection Checking on GPUs for Large-Scale Simulations," *ACM Transactions on Modeling and Computer Simulation*, Vol. 23, No. 3, Article 19, July 2013. (SCI, EI)
- J42. Wu-Chun Chung, Yi-Hsiang Lin, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "SARIDS: A Self-Adaptive Resource Index and Discovery System," *International Journal of Ad Hoc and Ubiquitous Computing*, Vol. 10, No. 2, July 2012, pp. 74-83(10). (SCI, EI)
- J43. Po-Chi Shih, Kuo-Chan Huang, and **Yeh-Ching Chung**, "Improving Processor Allocation in Heterogeneous Computing Grid through Considering Both Speed Heterogeneity and Resource Fragmentation," *The Journal of Supercomputing*, Vol. 60, No. 3, May 2012, pp. 311-337. (SCI, EI)
- J44. Jiazheng Zhou, Shen-En Liu, and **Yeh-Ching Chung**, "A Hardware Supported Multicast Scheme Based on XY Routing for 2-D Mesh InfiniBand Networks," *The Journal of Supercomputing*, Vol. 59, No. 2, February 2012, pp. 901-919. (SCI, EI)
- J45. Jiazheng Zhou and **Yeh-Ching Chung**, "Tree-Turn Routing: An Efficient Deadlock-Free Routing Algorithm for Irregular Networks," *The Journal of Supercomputing*, Vol. 59, No. 2, February 2012, pp. 882-900. (SCI, EI)
- J46. Wei-Chung Shia, Tien-Hsiung Ku, Yu-Ming Tsao, Chien-Hsun Hsia, Yung-Ming Chang, Ching-Hui Huang, **Yeh-Ching Chung**, Shih-Lan Hsu, Kae-Woei Liang, and Fang-Rong Hsu, "Genetic Copy Number Variants in Myocardial Infarction Patients with Hyperlipidemia," *BMC Genomics*, Vol. 12 (Suppl 3):S23, November 2011. (SCI, MEDLINE)
- J47. I-Hsin Chung, Che-rung Lee, Jiazheng Zhou, and **Yeh-Ching Chung**, "Hierarchical Mapping for HPC Applications," *Parallel Processing Letter*, Vol. 21, No. 3, 2011, pp. 279-299. (SCI, EI)
- J48. Chin-Jung Hsu, Wu-Chun Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Cooperative Failure Detection in Multi-Overlay Environments," *Journal of Internet Technology*, Vol. 12, No. 2, March 2011, pp. 259-267. (SCI, EI)
- J49. Che-Lun Hung, Chihan Lee, Chun-Yuan Lin, Chih-Hung Chang, Yeh-Ching Chung, and Chuan Yi Tang, "Feature Amplified Voting Algorithm for Functional Analysis of Protein Superfamily," *BMC Genomics*, Vol. 11 (Suppl 3):S14, December 2010. (SCI, MEDLINE)
- J50. Wu-Chun Chung, Chin-Jung Hsu, Yi-Shiang Lin, Kuan-Chou Lai, and **Yeh-Ching Chung**, "G2G: A Meta-Grid Framework for the Convergence of P2P and Grids," *International Journal of Grid and High Performance Computing*, Vol. 2, No. 3, July-September 2010, pp. 1-16.
- J51. Yu-Wei Chan, Tsung-Hsuan Ho, and **Yeh-Ching Chung**, "Malugo: A Peer-to-Peer Storage System," *International Journal of Ad Hoc and Ubiquitous Computing*, Vol. 5, No. 4, April 2010, pp. 209-218. (SCI, EI)

- J52. Yu-Wei Chan, Chih-Han Lai, and **Yeh-Ching Chung**, "A Construction of Peer-to-Peer Streaming System Based on Flexible Locality-Aware Overlay Networks," *International Journal of Pervasive Computing and Communications*, Vol. 6, No. 1, January-March 2010, pp. 104-124.
- J53. Chun-Hsien Wu and **Yeh-Ching Chung**, "A Polygon Model for Wireless Sensor Network Deployment with Directional Sensing Areas," *Sensors*, Vol. 9, No. 12, December 2009, pp. 9998-10022. (SCI, EI, NSC 98-2218-E-007-005, MOEA 96-EC-17-A-04-S1-044)
- J54. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, "Adaptive Processor Allocation for Moldable Jobs in Computational Grid," *International Journal of Grid and High Performance Computing*, Vol. 1, No. 1, January-March 2009, pp. 10-21.
- J55. Chun-Hsien Wu, Kuo-Chuan Lee, and **Yeh-Ching Chung**, "A Delaunay Triangulation Based Method for Wireless Sensor Network Deployment," *Computer Communications*, Vol. 30, No. 15, October, 2007, pp. 2744-2752. (SCI, EI, NSC94-2213-E-007-037, MOEA 94-EC-17-A-04-S1-044)
- J56. Chun-Yuan Lin and **Yeh-Ching Chung**, "Data Distribution Schemes of Sparse Arrays on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 41, No. 1, July 2007, pp. 63-87. (SCI, EI, NSC90-2213-E-035-019)
- J57. Jiazheng Zhou, Xuan-Yi Lin, and **Yeh-Ching Chung**, "Hardware Supported Multicast for Fat-Tree-Based InfiniBand Networks," *The Journal of Supercomputing*, Vol. 40, No. 3, June 2007, pp. 333-352. (SCI, EI, NSC93-2213-E-007-100)
- J58. Chin-Feng Lin, **Yeh-Ching Chung**, and Don-Lin Yang, "TRLE - An Efficient Data Compression Scheme for Image Composition of Volume Rendering on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 39, No.3, March 2007, pp. 321-345. (SCI, EI)
- J59. Jenshiuh Liu, Zhi-Jian Lee, and **Yeh-Ching Chung**, "Dynamic Probabilistic Packet Marking for Efficient IP Traceback," *Computer Networks*, Vol. 51, March 2007, pp. 866-882. (SCI, EI)
- J60. Chun-Yuan Lin and **Yeh-Ching Chung**, "Efficient Data Distribution Schemes for Multi-Dimensional Sparse Arrays," *Journal of Information Science and Engineering*, Vol. 23, No. 1, January 2007, pp. 315-327. (SCI, EI, NSC91-2213-E-007-104)
- J61. Chi-Sheng Shih, Shiao-Li Tsao, **Yeh-Ching Chung**, and Shyh-In Hwang, "Toward HW/SW Integration: a Networked Embedded System Course in Taiwan," *ACM SIGBED Review*, Vol. 4, No. 1, January 2007, pp. 36-54.
- J62. Tai-Yi Huang, Kuang-Li Huang, and **Yeh-Ching Chung**, "Performance Analysis of Hard-Real_time Embedded Software," *International Journal on Embedded Systems*, Vol. 2, No. 3/4, 2006, pp. 209-221.
- J63. Min-Hsuan Fan Chua-Huang Huang, **Yeh-Ching Chung**, Jen-Shiuh Liu, and Jei-Zhii Lee, "A Programming Methodology for Designing Block Recursive Algorithms", *Journal of Information Science and Engineering*, Vol. 22, No. 1, January 2006, pp. 95-121. (SCI, EI)
- J64. Chun-Yuan Lin, **Yeh-Ching Chung**, and Jen-Shiuh Liu, "Efficient Data Distribution Schemes for EKMR-based Sparse Arrays on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 34, No. 3, December 2005, pp. 291-313. (SCI, EI, NSC91-2213-E-007-104)

- J65. Ching-Feng Lin, Shih-Kuan Liao, **Yeh-Ching Chung**, and Don-Lin Yang, "A Rotate-Tiling Image Composition Method for Parallel Volume Rendering on Distributed Memory Multicomputers," *Journal of Information Science and Engineering*, Vol. 20, No. 4, July 2004, pp. 643-664. (SCI, EI, NSC89-2213-E-035-032)
- J66. Shih-Kuan Liao, Jim Z.C. Lai, and **Yeh-Ching Chung**, "Time-Critical Rendering for Time-Varying Volume Data," *Computers & Graphics*, Vol. 28, No. 2, April 2004, pp. 279-288. (SCI, EI)
- J67. Chun-Yuan Lin, **Yeh-Ching Chung**, and Jen-Shiuh Liu, "Efficient Data Compression Methods for Multi-Dimensional Array Operations Based on the *EKMR* Scheme," *IEEE Transactions on Computers*, Vol. 52, No. 12, December 2003, pp. 1640-1646. (SCI, EI, NSC90-2213-E-035-019)
- J68. Chun-Yuan Lin, Jen-Shiuh Liu, and **Yeh-Ching Chung**, "Efficient Data Parallel Algorithms for Multi-Dimensional Array Operations Based on the *EKMR* Scheme for Distributed Memory Multicomputers," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 14, No. 7, July 2003, pp. 625-639. (SCI, EI, NSC89-2213-E-035-007)
- J69. Shih-Kuan Liao, Ching-Feng Lin, **Yeh-Ching Chung**, and Jim Z.C. Lai, "A Differential Volume Rendering Method with Second-Order-Difference for Time-Varying-Volume-Data," *International Journal of Visual Languages and Computing*, Vol. 14, No. 3, June 2003, pp. 233-254. (SCI, EI)
- J70. Ching-Feng Lin, **Yeh-Ching Chung**, and Don-Lin Yang, "Parallel Shear-Warp Factorization Volume Rendering Using Efficient 1-D and 2-D Partitioning Schemes on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 22, No. 3, July 2002, pp. 277-302. (SCI, EI, NSC89-2213-E-035-032)
- J71. Chun-Yuan Lin, Jen-Shiuh Liu, and **Yeh-Ching Chung**, "Efficient Representation Scheme for Multi-Dimensional Array Operations," *IEEE Transactions on Computers*, Vol. 51, No. 3, March 2002, pp. 327-345. (SCI, EI, NSC89-2213-E-035-007)
- J72. Shih-Kuan Liao, **Yeh-Ching Chung**, and Jim Z.C. Lai, "A Two-Level Differential Volume Rendering Method for Time-Varying-Volume-Data," *The Journal of WSCG*, Vol. 10, No. 1, February 2002, pp. 287-294.
- J73. Ching-Hsien Hsu, **Yeh-Ching Chung**, Don-Lin Yang, and Chyi-Ren Dow, "A Generalized Processor Mapping Technique for Array Redistribution," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 12, No. 7, July 2001, pp. 743-757. (SCI, EI, NSC88-2213-E-035-002)
- J74. William C. Chu, Don-Lin Yang, Jen-Chih Yu, and **Yeh-Ching Chung**, "UMPAL-An Unstructured Mesh Partitioner and Load Balancer on World Wide Web," *Journal of Information Science and Engineering*, Vol. 17, No. 4, July 2001, pp. 595-614. (SCI, EI, NSC87-2213-E-035-010)
- J75. Don-Lin Yang, Jen-Chih Yu, and **Yeh-Ching Chung**, "Efficient Compositing Methods for the Sort-Last-Sparse Parallel Volume Rendering System on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 18, No. 2, February 2001, pp. 201-220. (SCI, EI, NSC89-2213-E-035-032)
- J76. Ching-Hsien Hsu, Sheng-Wen Bai, **Yeh-Ching Chung**, and Chu-Sing Yang "A Generalized Basic-Cycle Calculation Method for Array Redistribution," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 11, No. 12, December 2000, pp. 1201-1216. (SCI, EI, NSC87-2213-E-035-011)

- J77. Don-Lin Yang, **Yeh-Ching Chung**, Chih-Chang Chen, and Ching-Jung Liao, "A Dynamic Diffusion Optimization Method for Irregular Finite Element Graph Partitioning," *The Journal of Supercomputing*, Vol. 17, No. 1, August 2000, pp. 91-110. (SCI, EI, NSC87-2213-E-035-010)
- J78. Ching-Hsien Hsu, **Yeh-Ching Chung**, and Chyi-Ren Dow, "Efficient Methods for Multidimensional Array Redistribution," *The Journal of Supercomputing*, Vol. 17, No. 1, August 2000, pp. 23-46. (SCI, EI, NSC87-2213-E-035-011)
- J79. Don-Lin Yang, Ming-Hui Liu, **Yeh-Ching Chung**, and Chia-Hsien Wen, "Hybrid Interpolations to Enhance 3D Medical Image Reconstruction," *Chinese Journal of Medical and Biological Engineering*, Vol. 20, No. 1, March 2000, pp. 7-12. (NSC89-2213-E-035-032)
- J80. **Yeh-Ching Chung**, Ching-Jung Liao, and Don-Lin Yang, "A Prefix Code Matching Parallel Load Balancing Method for Solution-Adaptive Unstructured Finite Element Models on Distributed Memory Multicomputers," *The Journal of Supercomputing*, Vol. 15, No. 1, January 2000, pp. 25-50. (SCI, EI, NSC87-2213-E-035-010)
- J81. Ching-Jung Liao and **Yeh-Ching Chung**, "Tree-Based Load Balancing Methods for Solution-Adaptive Finite Element Graphs on Distributed Memory Multicomputers," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 10, No. 4, April 1999, pp. 360-370. (SCI, EI, NCHC-86-08-021)
- J82. Ching-Jung Liao and **Yeh-Ching Chung**, "A Binomial Tree Based Parallel Load-Balancing Method for Solution-Adaptive Finite Element Graphs on Distributed Memory Multicomputers," *Parallel Computational Fluid Dynamics*, Elsevier Science B.V., edited by C. Lin, A. Ecer, J. Periaux, and N. Satofuka, 1999, pp. 361-368. (NSC87-2213-E-035-010)
- J83. Ching-Hsien Hsu and **Yeh-Ching Chung**, "Efficient Methods for $kr \rightarrow r$ and $r \rightarrow kr$ Array Redistribution," *The Journal of Supercomputing*, Vol. 12, No. 3, May 1998, pp. 253-276. (SCI, EI, NSC86-2213-E-035-23)
- J84. **Yeh-Ching Chung**, Ching-Hsien Hsu, and Sheng-Wen Bai, "A Basic-Cycle Calculation Technique for Efficient Dynamic Data Redistribution," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 9, No. 4, April 1998, pp. 359-377. (SCI, EI, NSC86-2213-E-035-23)
- J85. Don-Lin Yang, Ching-Jung Liao, and **Yeh-Ching Chung**, "Parallel Virtual Reality Rendering of CT Images in a Distributed Computing Environment," *Chinese Journal of Medical and Biological Engineering*, Vol. 17, No. 2, June 1997, pp. 83-89.
- J86. **Yeh-Ching Chung**, "A Parallel Run-Time Iterative Load Balancing Algorithms for Solution-Adaptive Finite Element Meshes on Hypercubes," *Journal of the Chinese Institute of Engineers*, Vol. 19, No. 3, May 1996, pp. 291-301. (SCI, EI, NSC84-408-E-305-005)
- J87. **Yeh-Ching Chung**, Chia-Cheng Liu, and J.-S. Liu, "A Duplication Heuristic for Static Scheduling of Tasks on Distributed Memory Multiprocessors," *Journal of the Chinese Institute of Engineers*, Vol. 18, No. 6, November 1995, pp. 845-855. (SCI, EI)
- J88. **Yeh-Ching Chung**, Yaa-Jyun Yeh, and J.-S. Liu, "A Parallel Dynamic Load Balancing Algorithm for Solution-Adaptive Finite Element Meshes on 2D-Tori," *Concurrency: Practice and Experience*, Vol. 7, No. 7, October 1995, pp. 615-631. (SCI, EI, NSC84-408-E-035-005)
- J89. **Yeh-Ching Chung**, Chia-Cheng Liu, and J.-S. Liu, "Applications and Performance

- Analysis of a Compile-Time Optimization Approach for List Scheduling Algorithms on Distributed Memory Multiprocessors," *Journal of Information Science and Engineering*, Vol. 11, No. 2, June, 1995, pp. 155-181. (SCI, EI)
- J90. **Yeh-Ching Chung**, Wu-Hsun Ho, and Chia-Cheng Liu, "PPT: A Parallel Programming Tool for Distributed Memory Multiprocessors," *Journal of the Chinese Institute of Engineers*, Vol. 18, No. 3, May 1995, pp. 365-378. (SCI, EI, NSC82-0113-E-035-044T)
- J91. **Yeh-Ching Chung** and Sanjay Ranka, "Mapping Finite Element Graphs onto Hypercubes," *The Journal of Supercomputing*, Vol. 6, No. 3, September 1992, pp.257-282. (SCI, EI)

Conference Papers

- C1. Yuxuan Liu, **Yeh-Ching Chung**, and Shuang Chen, "Barbell: An Extensible Generator of On-Demand Loads for Interactive Cloud Services," *International Symposium on Advanced Parallel Processing Technology (APPT)*, July 2026. (CCF C)
- C2. Haotian Ma, Wei-Chung Hsu, and **Yeh-Ching Chung**, "HoloGraph: Bridging the Throughput Gap in Heterogeneous Graph Pattern Matching via Workload-Aware Steering," *Proceedings of ACM International Conference on Supercomputing (ICS)*, July 2026, pp. 565-575. (CCF B)
- C3. Yuhao Liu, Kevin William, Feige Zhou, **Yeh-Ching Chung**, and Wei-Chung Hsu, "Adaptive Block Size Selection for Translating Triton Kernels to RVV," *Proceedings of ACM International Conference on Research in Adaptive and Convergent Systems (RACS)*, November 2025, Article No. 2, pp. 1 - 8.
- C4. YongGang Li, Shunrong Jiang, Yu Bao, Pengpeng Chen, Yong Zhou, and **Yeh-Ching Chung**, "Isolate and Detect the Untrusted Driver with a Virtual Box," *Proceedings of ACM Conference on Computer and Communications Security (CCS)*, October 2024, pp. 4584-4597. (**The Outstanding Paper Award**, CCF A)
- C5. YuAng Chen and **Yeh-Ching Chung**, "Rebo: Locality-Aware Graph Processing via Reordering and Blocking," *Proceedings of IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, May 2026, pp. 1037-1049. (CCF B)
- C6. Shihao Hong and **Yeh-Ching Chung**, "ReShare: a Resource-Efficient Weight Pattern Sharing Scheme for Memristive DNN Accelerators", *IEEE International Symposium on Circuits & Systems (ISCAS)*, May 2024. (CCF C)
- C7. Weibin Chen and **Yeh-Ching Chung**, "An Offline Profile-Guided Optimization Strategy for Function Reordering on Relational Databases," *Proceedings of IEEE International conference on Systems, Man, and Cybernetics (SMC)*, October 2023, pp. 2323-2333. (CCF C)
- C8. Ao Shan and **Yeh-Ching Chung**, "A Hybrid Model Based on Samples Difficulty for Imbalanced Data Classification," *Proceedings of International Conference on Artificial Neural Network (ICANN)*, September 2023, pp. 26-37. (CCF C)
- C9. YuAng Chen and **Yeh-Ching Chung**, "Connectivity-Aware Link Analysis for Skewed Graphs," *Proceedings of International Conference on Parallel Processing (ICPP)*, August 2023, pp. 482-491. (CCF B)
- C10. Longbin. Lai, Yufan. Yang, Zhibin Wang, Yuxuan Liu, Haotian Ma, Sijie Shen, Bingqing Lyu, Xiaoli Zhou, Wwnyuan Yu, Zhengping Qian, Chen Tian, Sheng Zhong,

- Yeh-Ching. Chung**, and Jingren Zhou, "GLogS: Interactive Graph Pattern Matching Query At Large Scale," *Proceedings of USENIX Annual Technical Conference (ATC)*, July 2023, pp. 53-70. (CCF A)
- C11. Shihao Hong and **Yeh-Ching Chung**, "CR2: A Compressed ReRAM-Based DNN Accelerator by Combining Computation and Read Operation," *Design Automation Conference (DAC)*, July 2023, Poster. (CCF A)
- C12. YongGang Li, Yu Bao, GuoYuan Lin, Yaowen Ma, and **Yeh-Ching Chung**, "You Can Read My Code, But You Can't Execute It," *IEEE Proceedings of Design, Automation and Test in Europe Conference (DATE)*, April 2023, pp. 214-226. (CCF B)
- C13. YongGang Li, **Yeh-Ching Chung**, Jinbiao Xin, Yu Bao, Yi Lu, and GuoYuan Lin, "MProbe: Make the Code Probing Meaningless," *Proceedings of ACM Annual Computer Security Applications Conference (ACSAC)*, December 2022, pp. 214-226. (CCF B)
- C14. Weibin Chen and **Yeh-Ching Chung**, "Profile-Guided Optimization for Function Reordering: A Reinforcement Learning Approach," *Proceedings of IEEE International conference on Systems, Man, and Cybernetics (SMC)*, October 2022, pp. 2326-2333. (CCF C)
- C15. YuAng Chen and **Yeh-Ching Chung**, "HiPa: Hierarchical Partitioning for Fast PageRank on Multicore Systems," *Proceedings of IEEE International conference on Parallel Processing (ICPP)*, Article No. 24, 2021, pp. 1-10. (CCF B)
- C16. YuAng Chen and **Yeh-Ching Chung**, "Corder: Cache-Aware Reordering for Optimizing Graph Analytics," *Proceedings of ACM International conference on Principles and Practice of Parallel Programming (PPoPP)*, 2021, pp.472-473. (CCF A)
- C17. Guochao Xie and **Yeh-Ching Chung**, "Bucket-Based Expiration Algorithm: Improving Eviction Efficiency for In-Memory Key-Value Database," *Proceedings of ACM International Symposium on Memory Systems (MEMSYS)*, September 2020, pp.248-259.
- C18. Yu-Shiang Lin, Chun-Yuan Lin, Che-Rung Lee, and **Yeh-Ching Chung**, "qCUDA: GPGPU Virtualization for High Bandwidth Efficiency," *Proceedings of IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, December 2019, pp. 95-102.
- C19. Tse-Chuan Hsu, Terng-Yin Hsu, Hongji Yang, and **Yeh-Ching Chung**, "An IoT-Based Cloud-Fog Computing Platform for Creative Service Process," *IEEE Proceedings of International Conference on Computational Science and Computational Intelligence (CSCI)*, December 2017, pp. 1383-1388
- C20. Shih-Chun Huang, Bing-Liang Chen, Yu-Cing Luo, **Yeh-Ching Chung**, and Jerry Chou, "Application-Aware Traffic Redirection: A Mobile Edge Computing Implementation Toward Future 5G Networks," *Proceedings of IEEE International Symposiums on Cloud and Service Computing (SC2)*, December 2017, pp. 17-23. **(The Outstanding Paper Award)**
- C21. Bing-Liang Chen, Shih-Chun Huang, Yu-Cing Luo, **Yeh-Ching Chung**, and Jerry Chou, "A Dynamic Module Deployment Framework for M2M Platforms," *Proceedings of IEEE International Symposiums on Cloud and Service Computing*, December 2017, pp. 194-200.

- C22. Lidong Zhang, Yongwei Wu, Ruini Xue, Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung**, "HybridFS - A High Performance and Balanced File System Framework with Multiple Distributed File Systems," *Proceedings of IEEE Computer Software and Applications Conference (COMPSAC)*, July 2017, pp. 796-805.
- C23. Chun-Feng Wu, Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung**, "File Placement Mechanisms for Improving Write Throughputs of Cloud Storage Services Based on Ceph and HDFS," *Proceedings of IEEE International Conference on Applied System Innovation (ICASI)*, May 2017, pp. 1725-1728. (**The First Prize Paper Award**)
- C24. Jiayuan Zhang, Yongwei Wu, and **Yeh-Ching Chung**, "PROAR – A Weak Consistency Model for Ceph," *IEEE Proceedings of International Conference on Parallel and Distributed Systems (ICPADS)*, December 2016, pp. 348-353.
- C25. Hojjat Baghban, Mahdis Moradi, ching-Hsien Hsu, Jerry Chou, and **Yeh-Ching Chung**, "Byzantine Fault Tolerant Optimization in Federated Cloud Computing," *IEEE International Symposiums on Cloud and Service Computing (SC2)*, December 2016. (**The Best Poster Award**)
- C26. Moise W. Convolbo, Jerry Chou and **Yeh-Ching Chung**, "DRASH: A Data Replication-Aware Scheduler in Geo-Distributed Data Centers" *Proceedings of IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, December 2016, pp. 302-309.
- C27. Hao-Che Hsu, Chih-Wei Yeh, Shih-Hao Hung, Wei-Ching Hsu, Chung-Ta King, and **Yeh-Ching Chung**, "HSAemu 2.0 – Full System Emulation for HSA Platforms with Soft-MMU," *Proceedings of ACM International Conference on Research in Adaptive and Convergent Systems (RACS)*, October 2016, pp.230-236.
- C28. Hojjat Baghban, Jerry Chou, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "Modeling of Resource Granularity and Utilization with Virtual Machine Splitting," *Proceedings of IEEE International Conference on Big Data Intelligence and Computing (DataCom)*, August 2016, pp. 769-772.
- C29. Mu-Han Huang, Yu-Cing Luo, Chen-Nien Mao, Bing-Liang Chen, Shih-Chun Huang, Jerry Chou, Shun-Ren Yang, **Yeh-Ching Chung**, and Cheng-Hsin Hsu, "Performance Evaluations of Cloud Radio Access Networks," *Proceedings of EAI International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness (QSHINE)*, July 2016, pp 312-321.
- C30. Yu-Ju Huang, Hsuan-Heng Wu, **Yeh-Ching Chung**, and Wei-Chung Hsu, "Building a KVM-Based Hypervisor for a Heterogeneous System Architecture Compliant System," *Proceedings of the 12th ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments (VEE)*, April 2016, pp. 3-15. (**The Outstanding Paper Award**)
- C31. Lyu-Wei Wang, Shih-Chang Chen, Wenguang Chen, Hung-Chang Hsiao and **Yeh-Ching Chung**, "BiFennel: Fast Bipartite Graph Partitioning Algorithm for Big Data," *IEEE International Conference on Big Data Intelligence and Computing (DataCom)*, December 2015.
- C32. Yi-Lan Lin, Wu-Chun Chung, Cheng-Hsin Hsu, and **Yeh-Ching Chung**, "Community-Based M2M Framework using Smart/HetNet Gateways for Internet of Things," *Proceedings of IEEE International Conference on Internet of Things (iThings)*, December 2015, pp. 688-695.

- C33. Chen-Ning Mao, Mu-Han Huang, Satyajit Padhy, Shu-Ting Wang, Wu-Chun Chung, **Yeh-Ching Chung** and Cheng-Hsin Hsu, "Minimizing Latency of Real-Time Container Cloud for Software Radio Access Networks," *Proceedings of IEEE Workshop on QoS Assurance in the Cloud (QAC)*, December 2015, pp. 611-616.
- C34. Jing Wang, Wenqi Wang, Ming Zhao, Zhigang Tian, Shih-Chang Chen, **Yeh-Ching Chung**, Xing Cheng, and Yida Xu, "Evaluation of Inter-Cell Interference Coordination with CAP model," *Proceedings of International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness (QSHINE)*, August 2015, pp. 160-165.
- C35. Shu-Ming Chang, Chun-Feng Wu, Jiazheng Zhou, Wenguang Chen, Ching-Hsien Hsu, Hung-Chang Hsiao, and **Yeh-Ching Chung**, "Adaptive Fusion SQL Engine on Hadoop Framework," *Proceedings of International Conference on Advances in Big Data Analytics (ABDA)*, July 2015, pp. 27-33.
- C36. Su-Shien Ho, Jiazheng Zhou, Ching-Hsien Hsu, Hung-Chang Hsiao, Wenguang Chen, and **Yeh-Ching Chung**, "Distributed Metaserver Mechanism and Recovery Mechanism Support in Quantcast File System," *Proceedings of IEEE Computer Software and Applications Conference (COMPSAC)*, July 2015, pp. 758-963.
- C37. Jiun-Hung Ding, Bai-Cheng Jeng, Shih-Hao Hung, Wei-Chung Hsu, and **Yeh-Ching Chung**, "HSAemu – A Full System Emulator for HSA Platforms", *Proceedings of ACM International Conference on Hardware/Software Codesign and System Synthesis (CODES + ISSS)*, Article 26, October 2014.
- C38. R.C.Hsu, Shih-Chang.Chen, Ching-Hsien Hsu and **Yeh-Ching Chung**, "Corelation Aware Technique for SQL to NoSQL Transformation", *Proceedings of IEEE International Conference on Ubi-Media Computing*, July 2014, pp 43-46.
- C39. Wu-Chun Chung, Po-Chi Shih, Kuan-Chou Lai, Kuan-Ching Li, Che-Rung Lee, Jerry Chou, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "Taiwan UniCloud: A Cloud Testbed with Collaborative Cloud Services," *Proceedings of IEEE International Conference on Cloud Engineering (IC2E)*, March 2014, pp. 107-116.
- C40. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "Dynamic Data Partitioning and Virtual Machine Mapping: Efficient Data Intensive Computation with MapReduce," *Proceedings of IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2013, pp. 220-223.
- C41. Kenn Slagter, Ching-Hsien Hsu, **Yeh-Ching Chung** and Jong Hyuk Park, "Network-Aware Multiway Join for MapReduce", *Proceedings of the 8th International Grid and Pervasive Computing (GPC 2013)*, LNCS 7861, May 2013, pp. 73-80.
- C42. Chia-Hao Tsai, Jerry Chou, and **Yeh-Ching Chung**, "Value-Based Tiering Management on Heterogeneous Block-Level Storage System," *Proceedings of IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, Taipei, Taiwan, December 2012, pp. 393-398.
- C43. Guillermo A. Perez, Chung-Min Kao, Wei-Chung Hsu, and **Yeh-Ching Chung**, "A Hybrid Just-In-Time Compiler for Android," *Proceedings of ACM International Conference on Compilers, Architectures, and Synthesis of Embedded Systems (CASES)*, Tampere, Finland, October 2012, pp. 41-50.
- C44. Jiun-Hung Ding, Chang-Jung Lin, Ping-Hao Chang, Chieh-Hao Tsang, Wei-Chung Hsu, **Yeh-Ching Chung**, "ARMvisor: System Virtualization for ARM", *Linux Symposium*, Ottawa, Canada, July 2012, pp. 95-110.

- C45. Che-Rung Lee, Shih Hsiang Lo, Nan-Hsi Chen, **Yeh-Ching Chung**, and I-Hsin Chung, "GPU Performance Enhancement via Communication Cost Reduction: Case Studies of Radix Sort and WSN Relay Node Placement Problem," *Proceedings of IEEE/ACM International Symposium on Cluster, Cloud, and Grid Computing (CCGrid)*, Ottawa, Canada, May 2012, pp. 132-139.
- C46. Ding-Yong Hong, Chun-Chen Hsu, Pen-Chung Yew, Jan-Jan Wu, Wei-Chung Hsu, Pangfeng Liu, Chien-Min Wang, and **Yeh-Ching Chung**, "HQEMU: A Multi-Threaded and Retargetable Dynamic Binary Translator on Multicores," *Proceedings of ACM International Conference on Code Generation and Optimization (CGO)*, San Jose, USA, March 2012, pp. 104-113.
- C47. Jiun-Hung Ding, Po-Chun Chang, Wei-Chung Hsu, and **Yeh-Ching Chung**, "PQEMU: A Parallel System Emulator Based on QEMU," *IEEE Proceedings of International Conference on Parallel and Distributed Systems (ICPADS)*, Tainan, Taiwan, December 2011. (**The best paper award**)
- C48. Ssu-Hsuan Lu, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Design and Analysis of Arrangement Graph-based Overlay Systems for Information Sharing," *Proceedings of the 3rd IEEE International Workshop on Management of Emerging Networks and Services (IEEE MENS 2011)* in conjunction with IEEE GLOBECOM 2011, 5-9 December, Houston, Texas, USA, pp. 694-698.
- C49. Yu-Wei Chan, Ronald Y. Chang, Feng-Tsun Chien, Min-Kuan Chang, and **Yeh-Ching Chung**, "Coalitional Game Formulation for Multi-Channel Cooperative Cognitive Radio Networks," *Proceedings of 2011 IEEE International Workshop on Recent Advances in Cognitive Communications and Networking (RACCN)* in conjunction with IEEE GLOBECOM 2011, 5-9 December, Houston, Texas, USA, pp. 1010-1015.
- C50. Chih-Sheng Wang, Guillermo A. Perez, **Yeh-Ching Chung**, Wei-Chung Hsu, Wei-Kuan Shih, and Hong-Rong Hsu, "A Method-Based Ahead-of-Time Compiler for Android Applications," *Proceedings of ACM International Conference on Compilers, Architectures, and Synthesis of Embedded Systems (CASES)*, Taipei, Taiwan, October 2011.
- C51. Xuna-Yi Lin, Kuan-Chou lai, Shau-Yin Tseng, Kuan-Ching Li, and **Yeh-Ching Chung**, "An Efficient Programming Paradigm for Shared-Memory Master-Worker Video Decoding on TILE64 Many-Core Platform," *IEEE Proceedings of International Conference on Parallel Processing (ICPP)*, Taiwan, September 2011, pp. 414-423.
- C52. I-Hsin Chung, Che-Rung Lee, Jiazheng Zhou, and **Yeh-Ching Chung**, "Scalable Communication-Aware Task Mapping Algorithms for Interconnected Multicore Systems," *IEEE Proceedings of International Conference on High Performance Computing and Communication (HPCC)*, Banff, Canada, September 2011, pp. 759-764.
- C53. Shih-Hsiang Lo, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung**, "A Parallel Rectangle Intersection Algorithm on GPU+CPU," *Proceedings of IEEE/ACM International Symposium on Cluster, Cloud, and Grid Computing (CCGrid)*, Newport Beach USA, May 2011, pp. 43-52.
- C54. Po-Chi Shih, Kuo-Chan Huang, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung**, "TLPA: A Processor Allocation Technique for Heterogeneous Multi-Cluster Environments," *Proceedings of IEEE/ACM International Symposium on Cluster, Cloud, and Grid Computing (CCGrid)*, Newport Beach USA, May 2011, pp. 614-615.

- C55. Chin-Jung Hsu, Wu-Chun Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "A Novel Approach for Cooperative Overlay-Maintenance in Multi-Overlay Environments," *Proceeding of IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, Indianapolis, USA, November 2010, pp. 81-88.
- C56. Chun-Hsien Wu and **Yeh-Ching Chung**, "A Tiling-Based Approach for Directional Sensor Network Deployment," *Proceeding of IEEE SENSORS Conference*, Waikoloa, Hawaii, USA, November 2010, pp. 1358 - 1363.
- C57. **Yeh-Ching Chung**, Po-Chi Shih, Kuan-Ching Li, Chao-Tung Yang, Ching-Hsien Hsu, Fang-Rong Hsu, Don-Lin Yang, Chia-Hsien Wen, Chuang-Chien Chiu, "Medicare-Grid: new trends on the development of E-Health System based on Grid Technology," *Proceedings of World Computer Congress 2010, International Federation for Information Processing*, Brisbane, Australia, September 2010, pp. 148-159.
- C58. Shih-Hsiang Lo and **Yeh-Ching Chung**, "A Scalable Data Distribution Management Approach," *ACM Summer Computer Simulation Conference (SCSC)*, Ottawa, Canada, July 2010, pp. 570-575.
- C59. Ssu-Hsuan Lu, Kuan-Chou Lai, Don-Lin Yang, Kuan-Ching Li, and **Yeh-Ching Chung**, "Pervasive Health Service System: insights on the development of a Grid-based personal health service system," *Proceedings of the 12th International Conference on e-Health Networking, Application & Services (Healthcom)*, Lyon, France, July 2010, pp. 61-67.
- C60. Xuan-Yi Lin, Chung-Yu Huang, Pei-Man Yang, Tai-Wen Lung, Shau-Yin Tseng and **Yeh-Ching Chung**, "Parallelization of Motion JPEG Decoder on TILE64 Many-core Platform", *Proceedings of the 2nd Russia-Taiwan Symposium on Methods and Tools of Parallel Programming Multicomputers (MTPP)*, LNCS, Vladivostok, Russia, May 2010, pp. 59-68.
- C61. Meng-Hsuan Cheng, Kenn Slagter, Tai-Wen Lung, and **Yeh-Ching Chung**, "A VLIW-Based Post Compilation Framework for Multimedia Embedded DSPs with Hardware Specific Optimizations," *Proceedings of the 2nd Russia-Taiwan Symposium on Methods and Tools of Parallel Programming Multicomputers (MTPP)*, LNCS, Vladivostok, Russia, May 2010, pp. 51-58.
- C62. Ding-Yong Hong, Fang-Ping Pai, Shih-Hsiang Lo, and **Yeh-Ching Chung**, "A Scalable HLA RTI System Based on Multiple-FedServ Architecture," *Proceedings of IEEE International Conference on Computer Modeling and Simulation (UKSIM)*, Cambridge, UK, March 2010, pp. 527-532.
- C63. Shih-Hsiang Lo, Fang-Ping Pai and **Yeh-Ching Chung**, "Offloading Region Matching of Data Distribution Management with CUDA," *Proceedings of IEEE International Conference on Intelligent Systems, Modeling and Simulation (ISMS2010)*, Liverpool, UK, January 2010, pp. 306-311.
- C64. Yi-Hsiang Lin, Wu-Chun Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "SARIDS: A Self-Adaptive Resource Index and Discovery System," *Proceedings of IEEE International Symposium on Pervasive Systems, Algorithms, and Networks (I-SPAN)*, Kaohsiung, Taiwan, December 2009, pp. 521-526.
- C65. Po-Chi Shih, Kuo-Chan Huang, and **Yeh-Ching Chung**, "Improving Processor Allocation in Heterogeneous Computing Grid through Considering Both Speed Heterogeneity and Resource Fragmentation," *Proceedings of IEEE International*

Symposium on Pervasive Systems, Algorithms, and Networks (I-SPAN), Kaohsiung, Taiwan, December 2009, pp. 539-544.

- C66. Che-Lun Hung, Chun-Yuan Lin, **Yeh-Ching Chung**, and Chuan Yi Tang, "A Parallel Algorithm for Three-Profile Alignment Method," *Proceedings of IEEE International Joint Conference on Bioinformatics, Systems Biology and Intelligent Computing (IJCBS 2009)*, Shanghai, China, August 2009, pp. 153-159.
- C67. Yu-Hsun Lin, Xuan-Yi Lin, and **Yeh-Ching Chung**, "Reducing Leakage Power of JPEG Image on Asymmetric SRAM", *Proceedings of IEEE International Conference on Embedded and Ubiquitous Computing (EUC-09)*, Vancouver, Canada, August 2009, Vol. 2, pp. 374-381.
- C68. Jiazheng Zhou, Shen-En Liu, and **Yeh-Ching Chung**, "Hardware Supported Multicast in 2-D Mesh InfiniBand Networks," *Proceedings of IEEE International Conference on Computational Science and Engineering (CSE-09)*, Vancouver, Canada, August 2009, Vol. 1, pp. 232-239.
- C69. Wu-Chun Chung, Chin-Jung Hsu, Yi-Shiang Lin, Kuan-Chou Lai, and **Yeh-Ching Chung**, "G2G: A Meta-Grid Framework for the Convergence of P2P and Grids," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS), Geneva, Switzerland, May 2009, pp. 131-141.
- C70. Che-Lun Hung, Chun-Yuan Lin, and **Yeh-Ching Chung**, "CORAL-T: Heuristic CODing Region ALignment Method for Three Genome Sequences," Accepted by International Conference on Bioinformatics and Systems Biology (BSB 2009), Leipzig, Germany, March 2009
- C71. Shih-Hsiang Lo, Cheng-An Chiu, Ding-Yong Hong, Fang-Ping Pai and **Yeh-Ching Chung**, "MGRID: A Modifiable-Grid Region Matching Approach for DDM in the HLA RTI," *Proceedings of ACM High Performance Computing Symposium of 2009 Spring Simulation Multiconference*, San-Diego USA, March 2009, No. 114.
- C72. Tsung-Hsuan Ho, Yu-Wei Chan, and **Yeh-Ching Chung**, "Malugo: A Peer-to-Peer Storage System," Accepted by International Conference on High Performance Computing and Grid in Asia Pacific Region (HPC Asia), Taiwan, March 2009.
- C73. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, "Using Moldability to Improve Scheduling Performance of Parallel Jobs on Computational Grid," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS), China, May 2008, pp. 116-127.
- C74. Chih-Han Lai, Yu-Wei Chan, and **Yeh-Ching Chung**, "A Construction of Peer-to-Peer Streaming System Based on Flexible Locality-Aware Overlay Networks," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS), China, May 2008, pp. 296-307.
- C75. Tsung-Han Lin, Tsung-Hsuan Ho, Yu-Wei Chan, and **Yeh-Ching Chung**, "M-Ring: A Distributed, Self-Organize, Load Balancing Communication Method on Super Peer Network," *Proceedings of IEEE International Symposium on Parallel Architecture, Algorithms and Network (I-SPAN)*, Sydney, Australia, May 2008, pp. 59-64.

- C76. Po-Chi Shih, Hsi-Min Chen, **Yeh-Ching Chung**, Chien-Min Wang, Ruay-Shiung Chang, Ching-Hsien Hsu, Kuo-Chan Huang, and Chao-Tung Yang, "Middleware of Taiwan UniGrid," *Proceedings of ACM Symposium on Applied Computing (SAC)*, Brazil, March 2008, pp. 489-493.
- C77. Che-Lun Hung, Chun-Yuan Lin, **Yeh-Ching Chung**, and Chuan Yi Tang, "Introducing variable gap penalties into three-sequence alignment for protein sequences," *Proceedings of IEEE International Workshop on BioComputing (BioCom)*, Okinawa Japan, March 2008, pp. 726-731.
- C78. Chun-Yuan Lin, Chen-Tai Huang, **Yeh-Ching Chung**, and Chuan-Yi Tang, "Efficient Parallel Algorithm for Optimal Three-Sequences Alignment," *IEEE Proceedings of International Conference on Parallel Processing (ICPP 2007)*, China, September 2007.
- C79. Chih-Hsueh Yang, PeiZong Lee, and **Yeh-Ching Chung**, "Improving Static Task Scheduling in Heterogeneous and Homogeneous Computing Systems," *IEEE Proceedings of International Conference on Parallel Processing (ICPP)*, China, September 2007 (CD-ROM).
- C80. Shih-Hsiang Lo, Chun-Hsien Wu, and **Yeh-Ching Chung**, "Cooperative Localization with Pre-Knowledge using Bayesian Network for Wireless Sensor Networks," *IEEE Proceedings of ICPP Workshops on Ubiquitous Computing for Parallel and Distributed Systems (uPADS07)*, China, September 2007. (NSC 95-2221-E-007-018, MOEA 95-EC-17-A-04-S1-044)
- C81. Jer-Yu Hsu, Yan-Zu Wu, Xuan-Yi Lin, and **Yeh-Ching Chung**, "SCRF – A Hybrid Register File Architecture," *Proceedings of PaCT Russian-Taiwan Symposium on Methods and Tools of Parallel Programming Multicomputers*, Lecture Notes in Computer Science (LNCS), Russia, September 2007. (MOEA 95-EC-17-A-01-S1-038)
- C82. Chun-Hsien Wu and **Yeh-Ching Chung**, "Heterogeneous Wireless Sensor Network Deployment and Topology Control Based on Irregular Sensor Model," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS) 4459, France, May 2007, pp. 78-88. (SCI, EI, NSC 95-2221-E-007-018, MOEA 95-EC-17-A-04-S1-044)
- C83. Meng-Ru Lin, Ssu-Hsuan Lu, Tsung-Hsuan Ho, Peter Lin, and **Yeh-Ching Chung**, "CFR: A Peer-to-Peer Collaborative File Repository System," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS) 4459, France, May 2007, pp. 100-111. (SCI, EI, NSC 95-2752-E-007-004-PAE, NSC 94-2218-E-007-057)
- C84. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, "Towards Feasible and Effective Load Sharing in a Heterogeneous Computational Grid," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer Science (LNCS) 4459, France, May 2007, pp. 229-240. (SCI, EI, NSC 94-2218-E-007-057)
- C85. Shih-Hsiang Lo, Jiun-Hung Ding, Sheng-Je Hung, Jin-Wei Tang, Wei-Lun Tsai, and **Yeh-Ching Chung**, "SEMU: A Framework of Simulation Environment for Wireless Sensor Networks with co-simulation model," *Proceedings of International Conference on Grid and Pervasive Computing (GPC)*, Lecture Notes in Computer

Science (LNCS) 4459, France, May 2007, pp. 672-677. (SCI, EI, NSC 95-2221-E-007-018, MOEA 95-EC-17-A-04-S1-044)

- C86. Jiazheng Zhou, Xuan-Yi Lin, and **Yeh-Ching Chung**, "A Tree-Turn Model for Irregular Networks," *Proceedings of IEEE International Symposium on Network Computing and Applications* (NCA), USA, July 2006, pp. 11-18. (NSC94-2213-E-007-080)
- C87. Chun-Hsien Wu, Kuo-Chuan Lee, and **Yeh-Ching Chung**, "A Delaunay Triangulation Based Method for Wireless Sensor Network Deployment," *Proceedings of IEEE International Conference on Parallel and Distributed Systems* (ICPADS), USA, July 2006, pp. 253-260. (NSC94-2213-E-007-037)
- C88. Yang-Ping Cheng, Jiun-Hung Ding, Shih-Hsiang Lo, and **Yeh-Ching Chung**, "A Performance-Efficient Task Duplication-Based Scheduling Algorithm for Heterogeneous Computing," *Proceedings of International Conference on Parallel and Distributed Processing Techniques and Applications* (PDPTA), USA, June 2006. (NSC94-2213-E-007-080)
- C89. Ding-Yong Hong, Ching-Wen You, and **Yeh-Ching Chung**, "An Efficient MPI-IO for Noncontiguous Data Access over InfiniBand," *Proceedings of IEEE International Symposiums on Parallel Architecture, Algorithms, and Networks* (I-SPAN), USA, December 2005. (NSC94-2213-E-007-080)
- C90. Hsien-Tin Kuo, Jiazheng Zhou, Xuan-Yi Lin, and **Yeh-Ching Chung**, "An Efficient and Low Overhead Random Forwarding Table Construction Method for Deadlock-Free Routing Algorithms," *Proceedings of IASTED International Conference on Parallel and Distributed Computing and Systems* (PDCS), USA, November 2005. (NSC94-2213-E-007-080)
- C91. Jiazheng Zhou, Xuan-Yi Lin, Chun-Hsien Wu, and **Yeh-Ching Chung**, "Multicast in Fat-Tree-Based InfiniBand Networks," *IEEE Proceedings of International Conference on Network Computing and Applications* (NCA), USA, July 2005, pp. 239-242. (NSC93-2213-E-007-100)
- C92. Chang-Yi Lin, Jiazheng Zhou, and **Yeh-Ching Chung**, "An Adaptive Migratory Home Protocol for Software DSM Systems," *Proceedings of Information Technology: Research and Education* (ITRE), Taiwan, June 2005, pp. 272-276.
- C93. Yau-Ming Sun, Chih-Hsueh Yang, **Yeh-Ching Chung**, and Tai-Yi Huang, "An Efficient Deadlock-Free Tree-Based Routing Algorithm for Irregular Wormhole-Routed Networks Based on the Turn Model," *IEEE Proceedings of International Conference on Parallel Processing* (ICPP 2004), Canada, August 2004, pp. 343-352.
- C94. Chun-Yuan Lin, **Yeh-Ching Chung**, and Jen-Shiuh Liu, "Performance Evaluation of Data Distributions with Load-Balancing for Sparse Arrays," *Proceedings of IEEE International Symposiums on Parallel Architecture, Algorithms, and Networks* (I-SPAN), Hong Kong, May 2004, pp. 207-212.
- C95. Xuan-Yi Lin, **Yeh-Ching Chung**, and Tai-Yi Huang, "A Multiple LID Routing Scheme for Fat-Tree-Based InfiniBand Networks," *Proceedings of IEEE International Parallel and Distributed Processing Symposiums* (CD-ROM), USA, April 2004.

- C96. Jen-Shiuh Liu, Z.-J. Lee, and **Yeh-Ching Chung**, "Efficient Dynamic Probabilistic Packet Marking for IP Traceback," *Proceedings of 11th IEEE International Conference on Network*, Australia, September 2003, pp. 475-480.
- C97. Tai-Yi Huang, Kuang-Li Huang, and **Yeh-Ching Chung**, "An Iterative Integer Linear Programming Method for Bounding Program Performance on Embedded Systems," *Proceedings of the International Computer Symposium*, Taiwan, December 2002, pp. 142-149.
- C98. Chun-Yuan Lin, **Yeh-Ching Chung**, and Jen-Shiuh Liu, "Efficient Data Compression Schemes for Multi-Dimensional Array Operations Based on the *EKMR* Scheme," *IEEE Proceedings of Cyber World*, Japan, November 2002, pp. 61-69. (NSC90-2213-E-035-019)
- C99. Chin-Feng Lin, **Yeh-Ching Chung**, and Don-Lin Yang, "TRLE - An Efficient Data Compression Scheme for Image Composition of Volume Rendering on Distributed Memory Multicomputers," *IEEE Proceedings of Cyber World*, Japan, November 2002, pp. 499-506. (NSC89-2213-E-035-032)
- C100. Chin-Feng Lin, **Yeh-Ching Chung**, and Don-Lin Yang, "An Efficient Parallel Optical Flow Volume Rendering Method on Distributed Memory Multicomputers," *Proceedings of IASTED International Conference on Visualization, Imaging, and Image Processing (VIIP 2002)*, Spain, September 2002, pp. 551-556. (NSC90-2213-E-035-021)
- C101. William C. Chu, Chih-Wei Lu, Chih-Hung Chang, **Yeh-Ching Chung**, Yueh-Min Huang, Baowen Xu, "Software Maintainability Improvement: Integrating Standards and Models," *Proceedings of IEEE International Computer Software and Applications Conference (COMPSAC)*, England, October 2002, pp. 697-702. (EI)
- C102. Min-Hsuan Fan, Chua-Huang Hunag, and **Yeh-Ching Chung**, "A Programming Methodology for Designing Parallel Prefix Algorithms on Cube Networks," *IEEE Proceedings of ICPP Workshops on Compile/Runtime Techniques for Parallel Computing*, Canada, August 2002, pp. 607-614.
- C103. Chun-Yuan Lin, **Yeh-Ching Chung**, and Jen-Shiuh Liu, "Data Distribution Schemes of Sparse Arrays on Distributed Memory Multicomputers," *IEEE Proceedings of ICPP Workshops on Compile/Runtime Techniques for Parallel Computing*, Canada, August 2002, pp. 551-558. (NSC90-2213-E-035-019)
- C104. Wei-Ren Lin, Min-Hsuan Fan, Chua-Huang Hunag, **Yeh-Ching Chung**, and De-Sheng Chen "Synthesizing VHDL Programs from Tensor Product Formulas," *Proceedings of International Conference on VLSI*, USA, June 2002, pp. 134-140.
- C105. Don-Lin Yang, Ching-Ting Pan, and **Yeh-Ching Chung**, "An Efficient Hash-Based Method for Discovering the Maximal Frequent Set," *Proceedings of IEEE International Computer Software and Applications Conference (COMPSAC)*, USA, October 2001, pp. 511-516. (EI)
- C106. Min-Hsuan Fan, Chua-Huang Hunag, **Yeh-Ching Chung**, Jen-Shiuh Liu, and Jei-Zhii Lee, "A Programming Methodology for Designing Parallel Prefix Algorithms," *IEEE Proceedings of International Conference on Parallel Processing (ICPP)*, Spain, September 2001, pp. 463-470.

- C107. Chin-Feng Lin, Don-Lin Yang, and **Yeh-Ching Chung**, "A Marching Voxels Method for Surface Rendering of Volume Data," *IEEE Proceedings of Computer Graphics International*, Hong Kong, July 2001, pp. 306-313. (NSC89-2213-E-035-032)
- C108. Ching-Feng Lin, Don-Lin Yang, and **Yeh-Ching Chung**, "A Rotate-Tiling Image Composition Method for Parallel Volume Rendering on Distributed Memory Multicomputers," *Proceedings of IEEE International Parallel and Distributed Processing Symposiums USA*, April 2001, (CD-ROM). (NSC89213-E-035-032)
- C109. Jen-Shiuh Liu, Jiun-Yuan Lin, and **Yeh-Ching Chung**, "Efficient Parallel Algorithms for Multi-Dimensional Matrix Operations," *Proceedings of IEEE International Symposium on Parallel Architectures, Algorithms, and Networks (I-SPAN)*, USA, December 2000, pp. 224-229. (NSC89-2213-E-035-007)
- C110. Chao-Jen Lee and **Yeh-Ching Chung**, "A Web-Based Parallel PDE Solver Generation System for Distributed Memory Computing Environments," *Proceedings of IEEE International Computer Software and Applications Conference (COMPSAC)*, Taiwan, October 2000, pp. 54-59. (NSC90-2213-E-035-074)
- C111. Chung-Ya Liao, Jen-Shiuh Liu, and **Yeh-Ching Chung**, "JPDCE: a Java Parallel and Distributed Computing Environment," *Proceedings of the Fifth Symposium on Computer and Communication Technology*, Taiwan, Oct. 2000, pp. 3B19-3B24. (NSC88-2213-E-035-027)
- C112. Ching-Hsien Hsu and **Yeh-Ching Chung**, "An Efficient Processor Mapping Technique for Generalized $kr \rightarrow r$ and $r \rightarrow kr$ Array Redistribution," *Proceedings of International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA)*, USA, June 2000. (NSC-88-2213-E-035-002)
- C113. Jen-Shiuh Liu, Jiun-Yuan Lin, and **Yeh-Ching Chung**, "Efficient Representation for Multi-Dimensional Matrix Operations," *Proceedings of The Sixth Workshop on Compiler Techniques for High-Persormance Computing*, Taiwan, March 2000, pp. 133-142. (NSC89-2213-E-035-007)
- C114. Chin-Feng Lin, Don-Lin Yang, and **Yeh-Ching Chung**, "Parallel Shear-Warp Factorization Volume Rendering Using Efficient 1-D and 2-D Partitioning Schemes on Distributed Memory Multicomputers," *Proceedings of IASTED International Conference on Parallel and Distributed Computing and Systems (PDCS)*, USA, November 1999, pp. 555-560.
- C115. Don-Lin Yang, Jen-Chih Yu, and **Yeh-Ching Chung**, "Efficient Compositing Methods for the Sort-Last-Sparse Parallel Volume Rendering System on Distributed Memory Multicomputers," *Proceedings of IEEE International Conference on Parallel Processing (ICPP)*, Japan, Sept. 1999, pp. 200-207.
- C116. Don-Lin Yang, Jen-Chih Yu, Chih-Chang Chen, and **Yeh-Ching Chung**, "UMPAL-An Unstructured Mesh Partitioner and Load Balancer on World Wide Web," *Proceedings of The Sixth National Computational Fluid Dynamics Conference*, Taiwan, Aug. 1999, pp. 136-143.
- C117. William C. Chu, Chih-Wei Lu, Chih-Hung Chang, Ching-Jung Liao, and **Yeh-Ching Chung**, "A Parallel Program Generation Environment for Solving PDEs on

Distributed Memory Computing Environments," *Proceedings of International Conference on Software Engineering and Knowledge Engineering*, USA, June 1999, pp. 142-147.

- C118. **Yeh-Ching Chung**, Chih-Chang Chen, Ching-Jung Liao, and Don-Lin Yang, "Performance Comparisons of Partitioning and Load-Balancing Methods for Unstructured Adaptive Meshes on Distributed Memory Multicomputers," *Proceedings of SIAM International Conference on Parallel Processing for Scientific Computing*, USA, March 1999. (NSC87-2213-E-035-010)
- C119. Ching-Hsien Hsu, **Yeh-Ching Chung**, and Chyi-Ren Dow, "A Generalized Processor Mapping Technique for Array Redistribution," *Proceedings of Workshop on Compiler Techniques for High-Performance Computing*, Taiwan, March 1999, pp. 83-92. (NSC88-2213-E-035-002)
- C120. Ching-Hsien Hsu, **Yeh-Ching Chung**, and Chyi-Ren Dow, "Packing/Unpacking Information for Efficient Generalized $kr \rightarrow r$ and $r \rightarrow kr$ Array Redistribution," *Proceedings of IEEE Symposium on the Frontiers of Massively Parallel Computation*, USA, Feb. 1999, pp.89-97. (NSC87-2213-E-035-011)
- C121. Don-Lin Yang, Ching-Feng Lin, and **Yeh-Ching Chung**, "An Efficient Parallel Volume Rendering Algorithm Based on Shear-Warp Factorization on Distributed Memory Multicomputers," *Proceedings of International Computer Symposiums on Computer Graphics and Virtual Reality (ICS)*, Taiwan, Dec. 1998, pp. 44-51.
- C122. J.-S. Liu and **Yeh-Ching Chung**, "Lottery Drawing," *Proceedings of International Computer Symposiums on Algorithms (ICS)*, Taiwan, Dec. 1998, pp. 75-82.
- C123. **Yeh-Ching Chung**, Ching-Hsien Hsu, Sheng-Wen Bai, and Chu-Shing Yang "A Generalized Basic-Cycle Calculation Technique for Array Redistribution," *Proceedings of IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, Taiwan, Dec. 1998, pp. 640-647. (NSC87-2213-E-035-011)
- C124. **Yeh-Ching Chung**, Ching-Jung Liao, Chih-Chang Chen, and Don-Lin Yang "A Dynamic Diffusion Optimization Method for Irregular Finite Element Graph Partitioning", *Proceedings of IASTED International Conference on Parallel and Distributed Computing and Networks (PDCN)*, Australia, Dec. 1998, pp. 423-428. (NSC87-2213-E-035-010)
- C125. **Yeh-Ching Chung** and Ching-Hsien Hsu, "Efficient Methods for Multidimensional Array Redistribution," *Proceedings of IEEE International Conference on Parallel Architectures and Compilation Techniques (PACT)*, France, Oct. 1998, pp. 410-417. (NSC87-2213-E-035-011)
- C126. **Yeh-Ching Chung** and Ching-Jung Liao, "A Prefix Code Matching Parallel Load Balancing Method for Solution-Adaptive Unstructured Finite Element Models on Distributed Memory Multicomputers," *Proceedings of IEEE International Conference on Parallel Processing (ICPP)*, USA, August 1998, pp. 510-517. (NSC87-2213-E-035-010)
- C127. **Yeh-Ching Chung** and Ching-Jung Liao, "Tree-Based Load Balancing Methods for Solution-Adaptive Unstructured Finite Element Models on Distributed Memory Multicomputers," *Lecture Notes in Computer Science 1457* (Proceedings of Irregular'98), USA, August 1998, pp. 92-103. (NCHC-86-08-021)

- C128. D.L. Yang, C.C. Liu, **Yeh-Ching Chung**, J.S. Liu, C.R. Dow, C.F. Lin, 1998, "A Multicomputer System on Campus Networks", *Proceedings of Workshop on Distributed System Technologies & Applications*, Taiwan, May 1998, pp. 275-279.
- C129. **Yeh-Ching Chung** and Ching-Hsien Hsu, "Efficient Methods for $kr \rightarrow r$ and $r \rightarrow kr$ Array Redistribution," *Proceedings of IEEE International Computer Software and Applications Conference (COMPSAC)*, USA, August 1997, pp. 82-87. (NSC86-2213-E-035-23)
- C130. **Yeh-Ching Chung** and Ching-Hsien Hsu, "Message Encoding Techniques for Efficient Array Redistribution," *Proceedings of IEEE International Conference on Parallel Processing (ICPP)*, USA, August 1997, pp. 150-153. (NSC86-2213-E-035-23)
- C131. **Yeh-Ching Chung**, Ching-Hsien Hsu, and Sheng-Wen Bai, "A Basic-Cycle Calculation Technique for Efficient Dynamic Data Redistribution," *Proceedings of International Computer Symposium on Distributed Systems, Software Engineering, and Database Systems (ICS)*, Taiwan, Dec. 1996, pp. 137-144. (NSC86-2213-E-035-23)
- C132. Ching-Jung Liao, Chin-Feng Lin, and **Yeh-Ching Chung**, "A Parallel Recursive H/V Partitioning Method for Mapping Unstructured Finite Element Graphs on Processor Meshes," *Proceedings of International Computer Symposium on Distributed Systems, Software Engineering, and Database Systems (ICS)*, Taiwan, Dec. 1996, pp. 232-239. (NCHC-86-08-021)
- C133. **Yeh-Ching Chung**, Shih-Jey Chen, and J.-S. Liu, "HCL - An Implementation of Message Passing Interface on Workstation Clusters," *Proceedings of the 1996 Workshop on Distributed System Technologies and Applications*, Taiwan, May 1996, pp. 110-116. (NSC85-2213-E-035-018)
- C134. Chia-Cheng Liu, Yue-Che Chuang, Ching-Jung Liao, and **Yeh-Ching Chung**, "A LogP Model Implementation of the Collective Communication of MPI on A Cluster of Workstations," *Proceedings of The Second Workshop on Compiler Techniques for High Performance Computing (CTHPC)*, Taiwan, March 1996, pp. 87-95. (NSC85-2213-E-035-018)
- C135. **Yeh-Ching Chung**, Ching-Jung Liao, Shih-Jey Chen, and Chia-Cheng Liu, "A Prototype of Message-Passing Compiler with MPI standard for a Cluster of Workstations," *Proceedings of The First Workshop on Compiler Techniques for High Performance Computing (CTHPC)*, Taiwan, February 1995, pp. 24-29.
- C136. **Yeh-Ching Chung**, Yaa-Jyun Yeh, and J.-S.Liu, "A Parallel Dynamic Load Balancing Algorithm for Solution-Adaptive Finite Element Meshes on 2D-Tori," *Proceedings of International Computer Symposiums (ICS)*, Taiwan, Dec. 1994, pp.967-972. (NSC84-408-E-035-005)
- C137. **Yeh-Ching Chung**, Yaa-Jyun Yeh, and Chia-Cheng Liu, "A Parallel Run-Time Iterative Load Balancing Algorithms for Solution-Adaptive Finite Element Meshes on Hypercubes," *Proceedings of IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, Taiwan, Dec. 1994, pp.210-215. (NSC84-408-E-035-005)
- C138. **Yeh-Ching Chung** and Sanjay Ranka, "A Compile-Time Optimization Approach for

Scheduling of Tasks on Distributed Memory Multiprocessors," *Proceedings of IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, Taiwan, Dec. 1993, pp.87-91.

- C139. **Yeh-Ching Chung** and Sanjay Ranka, "A Parallel Program Design Tool for Single Program Multiple Data Model on Distributed Memory Multiprocessors," *Proceedings of IEEE Hawaii International Conference on System Sciences*, USA, Jan. 1993, pp.433-434.
- C140. **Yeh-Ching Chung** and Sanjay Ranka, "Applications and Performance Analysis of A Compile-Time Optimization Approach for List Scheduling Algorithms on Distributed Memory Multiprocessors," *Proceedings of IEEE Supercomputing*, USA, Nov. 1992, pp. 512-521.
- C141. Roger Chen and **Yeh-Ching Chung**, "Embedding Networks with Ring Connections in Hypercube Machine," *Proceedings of IEEE International Conference on Parallel Processing (ICPP)*, Vol. III, USA, August 1990, pp. 327-334.
- C142. **Yeh-Ching Chung** and Sanjay Ranka, "Mapping Finite Element Graphs onto Hypercubes," *Proceedings of IEEE Symposium on the Frontiers of Massively Parallel Computation*, USA, May 1990, pp. 135-144.

Supervised Ph.D. Students

- Ching-Hsien Hsu, January 1999
Dissertation Title – Efficient Methods for Run-Time Array Redistribution
- Ching-Jung Liao, June 1999
Dissertation Title – Efficient Partitioning and Load Balancing Methods for Finite Element Graphs on Distributed Memory Multicomputers
- Chun-Yuan Lin, June 2003
Dissertation Title – Efficient Data Parallel Program Design for Multi-Dimensional Array Operations Based on the Extended Karnaugh Map Representation Scheme
- Chin-Feng Lin, January 2004
Dissertation Title – Efficient Parallel Volume Rendering Methods on Distributed Memory Multicomputers
- Shih-Kuan Liao, June 2004
Dissertation Title – Time Varying Volume Data Rendering
- Che-Lun Hung, March 2010
Dissertation Title – Efficient Algorithms for Triple-wise Alignment and its Applications
- Chun-Hsien Wu, June 2010
Dissertation Title – Deployment Strategies for Wireless Sensor Networks
- Jia-Zheng Zhou, March 2011
Dissertation Title – Routing in Interconnection Networks and Its Applications
- Po-Chi Shih, July 2012
Dissertation Title – Improving Parallel Job Scheduling Performance in Multi-Cluster through Look-Ahead Processor Allocation

- Shih-Hsiang Lo, March 2013
Dissertation Title – Optimization Techniques for Reducing Data Communication Costs on GPU Systems
 - Wu-Chun Chung, April 2013
Dissertation Title – Decentralized Management of Resource Information and Discovery in Large-Scale Distributed Computing
 - Ding-Yong Hong, April 2013
Dissertation Title – Efficient and Retargetable Dynamic Binary Translation
 - Xuan-Yi Lin, June 2013
Dissertation Title – Master-Worker Paradigm on TILE64 Many-Core Platform
 - Yu-Wei Chan, June 2013
Dissertation Title – Resource Sharing Strategies for Wide-Area Fusing Locality-Aware and Game Theoretic Approaches
 - Kenn Slagter, November 2013
Dissertation Title – Load Balancing and Data Placement in MapReduce
 - Ssu-Hsuan Lu, January 2014
Dissertation Title – A Self-Scaling Overlay Network Based on Arrangement Graph
 - Fang-Ping Bai, July 2016
Dissertation Title – An Ontology-Driven Multi-Agent Information Fusion Framework to Enhance Situation Awareness
 - Moïse Wendkuuni Convolbo, July 2017
Dissertation Title – Scheduling Techniques on Cloud Resources under Cost and Performance Requirements
 - Yu-Hsiang Lin, January 2018
Dissertation Title – A New API Remoting Policy for General-Purpose Computing on GPU Virtualization
 - YuAng Chen, May 2022
Dissertation Title – High-Performance Graph Processing on Multicore Systems
 - Ao Shan, November 2023
Dissertation Title – Imbalanced Data Classification Problems Based on Data Complexity
 - Weibin Chen, September 2024
Dissertation Title – Profile-Guided Optimization for Function Reordering: Algorithm and Application
 - Lele Li, December 2024
Dissertation Title – On the Co-optimization of Performance and Resource Efficiency for Big Data Analytics
 - Renwen Ma, January 2025 (Co-Supervised with Professor Kai Hwang)
Dissertation Title – Securing Federated Machine Learning Systems with Blockchain and Zero-Knowledge Proofs
 - Shihao Hong, May 2025
Dissertation Title – Hardware-Aware Redundancy Pruning for DNN Inference: From ReRAM PIM to GPU
-

Supervised Master Students

- Yaa-Jyun Yeh, June 1994
Thesis Title – Parallel Dynamic Load Balancing Algorithms for Solution-Adaptive Finite Element Graphs on Distributed Memory Multiprocessors
- Wu-Hsun Ho, June 1995
Thesis Title – PPT: A Parallel Programming Tool for Distributed Memory Multiprocessors
- Shih-Jey Chen, June 1996
Thesis Title – HCL: An Implementation of Message Passing Interface on Workstation Clusters
- Sheng-Wen Bai, June 1998
Thesis Title – A Generalized Basic-Cycle Calculation Technique for Efficient Array Redistribution
- Jen-Chih Yu, June 1999
Thesis Title – Efficient Compositing Methods for the Sort-Last-Sparse Parallel Volume Rendering System on Distributed Memory Multicomputers
- Chih-Chang Chen, June 1999
Thesis Title – A Dynamic Diffusion Optimization Method for Irregular Finite Element Graph Partitioning
- Chao-Jen Lee, June 2000
Thesis Title – A Web-Based Parallel PDE Solver Generation System for Distributed Memory Computing Environments
- Chun-Yuan Lin, June 2000
Thesis Title – Efficient Algorithms for Multi-Dimensional Matrix Operations
- Ching-Ting Pan, June 2001
Thesis Title – An Efficient Hash Based Method for Discovering the Maximal Frequent Set
- Su-Ming Yang, June 2001
Thesis Title – The Design and Implementation of an IAM2000s C Compiler
- Wu-Hong Chen, June 2001
Thesis Title – A Real-Time Flight Control Embedded System Based on uC/OS-II Kernel
- Ching-Jung Long, January 2002
Thesis Title – Efficient Load Balancing Methods for Cellular Mobile Networks
- Chun-Hsing Chang, June 2002
Thesis Title – A Configurable Hard Real-Time Operating System Based on uC/OS-II Kernel for Flight Controls
- Yuan-Chang Tsai, June 2002
Thesis Title – An Integrated Design for Automatic System Development Tool Kits Generation
- Chun-Hsien Wu, June 2002
Thesis Title – A Service Composition Framework for Service Discovery/Delivery Based on SLP
- Kuen-Lin Cheng, June 2002
Thesis Title – The Integration of GPS and GIS on PDAs

- Jen-Shen Lin, June 2002
Thesis Title – Image Compression by Adjacent Decomposed Bit-Plane Substitution
- Xuan-Yi Lin, June 2003
Thesis Title – An Efficient Communication Scheme for Fat-Tree Topology on InfiniBand Networks
- Yao-Ming Sun, July 2004
Thesis Title – An Efficient Deadlock-Free Tree-Based Routing Algorithm for Irregular Wormhole-Routed Networks Based on the Turn Model
- Jia-Zheng Zhou, July 2004
Thesis Title – Multicast in Fat-Tree-Based InfiniBand Networks
- Chang-Yi Lin, July 2004
Thesis Title – An Adaptive Migratory Home Protocol for Software Distributed Shared Memory Systems
- Yun-Chan Lee, July 2005
Thesis Title – A Hybrid Wavelet-Based Video Codec with Low Computational Complexity, Memory Efficiency, and Scalability
- Ssu-Chun Tseng, July 2005
Thesis Title – A Scalable and Robust Scheme Based on the Multiple Description Coding for Low Bit-Rate Speech Coders
- Tse-Ming Kao, July 2005
Thesis Title – A Multithreaded DSM System with Thread Migration for Memory Locality over InfiniBand
- Kuo-Chuan Lee, July 2005
Thesis Title – A Delaunay Triangulation Based Method for Wireless Sensor Network Deployment
- Hsien-Ting Kuo, July 2005
Thesis Title – An Efficient and Low Overhead Random Forwarding Table Construction Method for Deadlock-Free Routing Algorithms in InfiniBand Networks
- Ding-Yong Hong, July 2005
Thesis Title – An Efficient MPI-IO for Noncontiguous I/O over InfiniBand
- Shen-En Liu, July 2005
Thesis Title – Hardware Supported Multicast in Tow-Dimensional Mesh InfiniBand Networks
- Wei-Ching Kao, September 2005
Thesis Title – An embedded Print Server
- Yang-Ping Cheng, September 2005
Thesis Title – A Performance-Efficient Task Duplication- Based Scheduling Algorithm for Heterogeneous Computing
- Tse-Yu Hsu, June 2006
Thesis Title – A Register Allocation Algorithm for Shared and Clustered Hybrid Register File Organization
- Meng-Ru Lin, June 2006
Thesis Title – CFR: A Peer-to-Peer Collaborative File Repository System
- Ching-Wen Yu, June 2006

Thesis Title – SPE: A Programming Environment for Wireless Sensor Networks

- Yi-Kai Pong, June 2006
Thesis Title – Applying Curve Fitting Techniques to Construct the Synopsis of Data Streams
- Jiun-Hung Ding, June 2006
Thesis Title – SEMU: A Framework of Simulation Environment for Wireless Sensor Networks
- Shih-Hsiang Lo, June 2006
Thesis Title – Cooperative Localization with Pre-Knowledge using Bayesian Network for Wireless Sensor Networks
- Wen-Li Tang, June 2006
Thesis Title – A Fast, Flexible, and Precise Simulation Framework for Starfish DSP Architecture
- Chih-Han Lai, June 2007
Thesis Title – A Construction of Peer-to-Peer Streaming System Based on Flexible Locality-Aware Overlay Networks
- Tun-Yi Chang, June 2007
Thesis Title – Taking Advantage of Multi-Core Architecture in Disk Array
- Yu-Chen Wu, July 2007
Thesis Title – Dynamic Helper Threaded Prefetching on the Multi-Core Software RAID
- Chin-Wei Tang, July 2007
Thesis Title – A System-Level Simulation Framework for Performance Evaluation of Hardware-Based Network Intrusion Detection Systems
- Meng-Hsuan Cheng, July 2007
Thesis Title – A VLIW-Based Post Compilation Framework for Multimedia Embedded DSPs with Hardware Specific Optimizations
- Wei-Lun Tsai, July 2007
Thesis Title – A Tag-Based Overlay Network for Defending Against DDoS Attack
- Yan-Zu Wu, July 2007
Thesis Title – A Extended Shared and Clustered Register Files Organization – Wired SCRF Architecture
- Chung Han Lin, July 2007
Thesis Title – M-Ring: A Distributed, Self-Organize, Load Balancing Communication Method on Super Peer Network
- Sheng-Je Hung, July 2007
Thesis Title – A Pipeline and Macroblock-Level Parallelism Architecture Design for H.264/AVC Baseline Decoder
- Yi-Hsin Lin, July 2008
Thesis Title – Reduce Leakage Power of JPEG Image on Asymmetric SRAM
- Chung-Hsuan Ho, July 2008
Thesis Title – Malugo: A Peer-to-Peer Storage System
- Meng-Ju Hsieh, July 2008
Thesis Title – Minimize the Distance of Failure Location for Automatic Debugging Technique with Improved State-Coverage mechanism

- Chih-Chen Chen, July 2008
Thesis Title – Using parity cache to reduce overhead of small write requests under Linux software RAID-5 system
- Hao Yuan, July 2008
Thesis Title – Profile Based Trace Selection of Code Layout Optimizations in Open64 Compiler
- Cheng-Tin Tsai, July 2008
Thesis Title – A Parallel Algorithm of H.264/AVC Encoder on Multi-Core Systems
- Cheng-An Chiu, July 2008
Thesis Title – An Adaptive Method for Distributed Data Management in High Level Architecture
- Fu-Mao Chen, July 2008
Thesis Title – A Peer-to-Peer High Level Architecture
- Chih-Chien Chang, June 2009
Thesis Title – Helper Threaded Shadow Paging on Multi-Core Virtualization Processors
- Chin-Jung Hsu, June 2009
Thesis Title – Cooperative Maintenance in Multi-Overlay Environments
- Yi-Hsiang Lin, June 2009
Thesis Title – SARIDS : A Self-Adaptive Resource Index and Discovery System
- Ze-Chuan Yeh, June 2009
Thesis Title – A Power-Efficient and High-Throughput Deblocking Filter Hardware Architecture for H.264/AVC
- Chia-Dar Tsai, July 2009
Thesis Title – A Low-Power Hardware Design for H.264/AVC Baseline Decoder
- Chih-Yin Wu, July 2009
Thesis Title – Dryad: An Ahead of Time Compiler Optimization Framework for Android
- Min-Chuan Hung, July 2009
Thesis Title – A Time/Space Balanced Flash Translation Layer for Embedded Systems
- Shuo-Hung Chen, July 2010
Thesis Title – HW/SW Co-Design of High-Performance Multi-Core Vector Graphics System
- Wei-Chao Chang, July 2010
Thesis Title – A Locality-based Multicast Scheme on Structured Peer-to-peer Overlay for High Level Architecture
- Chia-Hong Chien, September 2010
Thesis Title – A Separate Code Cache Model for a Parallel Multi-Core System Emulator Based on QEMU
- Hsiang-Wei Sung, September 2010
Thesis Title – Processor virtualization and optimization for KVM-ARM
- Fan-Lei Liao, July 2011
Thesis Title – Memory Virtualization Design for KVM-ARM
- Tai-Wen Lung, July 2011
Thesis Title – Data Locality Optimizations for Multi-Level Caches in Java Multi-Core Compiler

- Chih-Sheng Wang, July 2011
Thesis Title – An Ahead-of-Time compilation and optimization framework based on LLVM for Google Android
- Moise Convolbo, July 2011
Thesis Title – Rsvg: Display Big Size Vector Map on Mobile Device
- Ya-Ting Chang, May 2012
Thesis Title – NUMA-Aware Scheduler for Asymmetric Multicore Processors
- Guillermo A. Perez, May 2012
Thesis Title – A Hybrid Just-In-Time Compiler for Android
- Ming-Hsuan Lee, June 2012
Thesis Title – The Scheduling and Timer Interrupt Mechanism for PQEMU
- Chao-Chu Wang, June 2012
Thesis Title – Implementation of x86 Atomic Instruction Emulation in PQEMU
- Jia-Wang Chung, June 2012
Thesis Title – Interrupt Controller Design Based on PQEMU
- Chang-Jung Lin, June 2012
Thesis Title – I/O Performance Optimization and Analysis for ARMvisor Using Virtio and Irqchip
- Yu-Ying Chen, June 2012
Thesis Title – Improve 3D Performance of Qemu-Spice
- Shih-Wei Li, June 2012
Thesis Title – Memory Virtualization for ARMvisor
- Hung-Ping Lin, June 2012
Thesis Title – Structured Data Processing on MapReduce in NoSQL Database
- Yi-Man Ma, July 2012
Thesis Title – InfiniBand Virtualization for Kernel-based Virtual Machine
- Zhoudong Guo, June 2013
Thesis Title – HSA Emulator Design Based on QEMU
- Sherif Ceesay, June 2013
Thesis Title – Moving Towards Pure ANSI SQL in NoSQL
- Chih-Hao Chang, July 2013
Thesis Title – Memory Virtualization on ARM Architecture with Virtualization Hardware Extension
- Chung-Min Kao, July 2013
Thesis Title – The LLVM Based GPU Compiler in Heterogeneous System Architecture Emulator: HTranslator
- Ping-Hao Chang, July 2013
Thesis Title – CPU Virtualization with ARM Virtualization Extension on ARMvisor
- Shu-Ming Chang, July 2014
Thesis Title – Adaptive Fusion SQL Engine on Hadoop Framework

- Ying-Ti Liao, July 2014
Thesis Title – Data Adapter for Querying and Transformation Between SQL and NoSQL Databases
- Su-Shien Ho, July 2014
Thesis Title – Distributed Metaserver Mechanism and Recovery Mechanism Support in Quanteast File System
- Yi-chen Yang, July 2014
Thesis Title – CSCJ: Star Join Processing on HBase Using Hadoop MapReduce
- Intan Maghfirah, July 2014
Thesis Title – Constraints Preserving in Schema Transformation to Enhance Database Migration from MySQL to HBase
- Ngo-Thi Nha, July 2014
Thesis Title – MTSH: Multi-Tenant Web-based Management System on SSDB and HDFS
- Kuo-Min Lin, July 2014
Thesis Title – A Compilation Framework for HSA
- Bai-Cheng Jeng, July 2014
Thesis Title – HSAemu Framework
- Che-Yang Kuo, July 2015
Thesis Title – Implementation of Image Feature Supports in HSAemu Framework
- Jui Hsiao, July 2015
Thesis Title – An OpenCL 2.0 Compilation Framework for HSA
- Fei Peng, July 2015
Thesis Title – OpenCL 2.0 Enabled HSA Hardware Emulation
- Yen-Yu Lin, July 2015
Thesis Title – Container-Based Automated Workflow System Supporting On-Demand Service Provision
- Hsing-Chang Chou, July 2015
Thesis Title – Container-Based Scale-Out Architecture for Cloud Storage Service
- Wei-Chih Sun, July 2015
Thesis Title – An OpenCL 2.0 Runtime Based on HSA Runtime
- Lyu-Wei Wang, July 2015
Thesis Title – BiFennel: Fast Graph Partitioning Algorithm for Big Data
- Chih-Hsien Chen, July 2015
Thesis Title – SQL/PCM Design and Implement on NoSQL Database
- Yi-Lan Lin, July 2015
Thesis Title – Community-Based M2M Framework Using Smart/HetNet GW
- Chun-Feng Wu, June 2016
Thesis Title – A Hybrid Mechanism to Improve Write Scenarios for Cloud Storage Services

- Mu-Han Huang, June 2016
Thesis Title – Application-Aware Resource Provisioning for On-Demand Container-Based Cloud Radio Access Network
- Chia-Hung Lu, June 2016
Thesis Title – Log-Based Synchronization Mechanism Between RDBMS and NoSQL Databases
- Luis Herrera, June 2016
Thesis Title – A KVM-Based GPGPU Virtualization Technique for Windows
- Hao-Che Hsu, June 2016
Thesis Title – An HSA Emulation Platform with SVM and Multiple Computing Devices Support
- Beibei Li, June 2016
Thesis Title – Container-Based Workflow System for Production Server Deployment
- Shih-Chun Huang, July 2017
Thesis Title – Application-Aware Offloading Mechanism for Mobile Edge Computing
- Yu-Chin Luo, July 2017
Thesis Title – Improve Resource Utilization of Cloud Radio Access Network Using Containerization and Functional Split Techniques
- Bing-Liang Chen, July 2017
Thesis Title – Dynamic Module Deployment Mechanism and Placement Optimization on M2M Platform
- Hung-Ying Tai, July 2017
Thesis Title – Crystal: An Extensible Framework for Profiling and Optimizing Large-Scale System Software
- Meng-Hsun Yang, July 2017
Thesis Title – Code Layout Optimization Applying Community Detection Algorithm
- Jian-Lin Chen, July 2017
Thesis Title – Function Reordering Optimization Based on Weighted Call Graph for Large-Scale Software in LLVM
- Wenyu Huang, July 2023
Thesis Title – ReKV: A Good Read Performance LSM-TreeKey-Value Store
- Yufan Yang, December 2023
Thesis Title – HWO: High-Order Based and WCO Guaranteed Query Plan Optimizer for Graph Pattern Matching
- Yihan Liu, December 2024
Thesis Title – V2M: Automatic Music Generation for a Given Video Based on Video Understanding and Emotion Detection using a Multimodal Transformer Model

Keynote/Tutorial Speech (since 2010)

- Keynote Speaker, IEEE Big Data Congress Taipei Satellite Session, Taipei, Taiwan, 2015

Title – SSBDS: A High-Performance Big Data Service for Research and Education

- HSA Tutorial Speaker, International Symposiums on Computer Architecture (ISCA), Minneapolis, USA, 2014

Title – HSA Runtime

- Keynote Speaker, China Conference on Cloud Computing, Guangzhou, China, 2013

Title – UniCloud: A Community/Hybrid Cloud Platform for Research and Education

- Keynote Speaker, International Conference on Computational Intelligence, Modeling and Simulation, Bali, Indonesia, 2010

Title – Computer Go Game

Conference Chairmanships (Since 2006)

- Advisory Committee Member of International Conference on Big Data Intelligence and Computing (DataCom) 2019, November 18-21, 2019
- General Co-Chair of International Conference on Big Data Intelligence and Computing (DataCom) 2015, December 19-21, 2015
- Program Co-Chair of IEEE International Conference on Cloud Computing and Big Data (CCBD), December 4 – 6, 2015
- Program Co-Chair of IEEE International Conference on Parallel and Distributed Systems (ICPADS), December 16 – 19, 2014
- General Co-Chair of IEEE International Conference on Cloud Computing Technology and Science (CloudCom), December 3-6, 2012
- Track Co-Chair of International Conference on Parallel Processing (ICPP), September 13-16, 2011
- Track Co-Chair of International Conference on Contemporary Computing (IC3), August 8-10, 2010
- General Co-Chair of International Symposium on Information Assurance and Security Management (IASM), December 14-16, 2009
- Track Co-Chair of International Symposium on Pervasive Systems, Algorithms, and Networks (I-SPAN), December 14-16, 2009
- Workshop Co-Chair of National Computer Symposium (NCS) 2009, November 27-28, 2009
- Track Co-Chair of International Conference on Contemporary Computing (IC3), August 17-19, 2009
- Program Co-Chair of International Symposium on Grid and Distributed Computing (GDC), December 13-15, 2008
- Award Chair of IEEE Asia-Pacific Services Computing Conference (APSCC), December 09-12, 2008
- Program Co-Chair of IEEE Asia-Pacific Computer Systems Architecture Conference (ACSAC), August 20-22, 2008
- Program Co-Chair of International Conference on Grid and Pervasive Computing (GPC)

2006), May 2-4, 2006

Program Committee memberships (Since 2007)

- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, May 28 - June 1, 2023
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, May 16-11, 2022
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, June 14-18, 2021
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, June 7-11, 2020
- IEEE International Conference on Communications (ICC-2018) SAC Symposium on Big Data Track, May 20-24, 2018
- IEEE International Global Communications Conference (Globecom-2017) SAC on Big Data Track, December 4-8, 2017
- IEEE International Conference on Communications (ICC-2017) SAC Symposium on Big Data Networking Track, May 21-25, 2017
- IEEE International Global Communications Conference (Globecom-2016) SAC on Big Data Track, December 4-8, 2016
- International Conference on Network and Parallel Computing (NPC-2016), October 28-29, 2016
- International Conference on Green and Pervasive, and Cloud Computing (GPC-2015), May 4-6, 2015
- IEEE International Conference on Cloud Engineering (IC2E-2015), March 9-12, 2015
- IEEE International Conference on Green Computing and Communications (GreenCom-2014), September 1-3, 2014
- IEEE International Conference on Embedded Computing (EmbeddedCom-2014), August 24-27, 2014
- International Conference on Green and Pervasive, and Cloud Computing (GPC-2014), May 26-28, 2014
- IEEE International Conference on Green Computing and Communications (GreenCom-2013), August 20-23, 2013
- International Conference on Smart Grids and Green IT Systems (SMARTGREENS-2013), May 9-10, 2013
- International Conference on Grid and Pervasive computing (GPC-2013), May 9-11, 2013
- IEEE Asia Pacific Cloud Computing Congress (APCloud-2012), November 14-17, 2012
- IEEE International Parallel & Distributed Processing Symposium (IPDPS-2012) May 21-25, 2012
- International Conference on Grid and Pervasive Computing (GPC-2012), May 11-13, 2012
- IEEE International Conference on High Performance Computing (HiPC-2011), December 19-22, 2011

- International Symposium on Grid and Distributed Computing (GDC-2011), December 8-10, 2011
- International Conference on Network and Parallel Computing (NPC-2011), October 21-23, 2011
- International Conference on Contemporary Computing (IC3-2011), August 8-10, 2011
- IEEE/ACM International Conference on Green Computing and Communications (GreenCom-2011), August 4-5, 2010.
- IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA-2011), May 26-28, 2011.
- International Conference on Grid and Pervasive Computing (GPC-2011), May 11-13, 2011
- International Symposium on Grid and Distributed Computing (GDC-2010), November 11-13, 2010
- IEEE/ACM International Conference on Green Computing and Communications (GreenCom-2010), October 30-November 1, 2010.
- International Conference on Advances in P2P Systems (AP2PS-2010), October 25 - 30, 2010.
- IEEE International Conference on Embedded Software and Systems (ICESS-2010) June 29-July 1, 2010.
- International Conference on Grid and Pervasive Computing (GPC-2010), May 10-13, 2010
- International Conference on Frontier of Computer Science and Technology (FCST-2009), December 17-19, 2009
- IEEE International Conference on High Performance Computing (HiPC-2009), December 16-19, 2009
- International Symposium on Grid and Distributed Computing (GDC-2009), December 10-12, 2009
- International Conference on Embedded and Multimedia Computing (EM-Com-2009), December 10 - 12, 2009
- International Conference on Computer Science and its Applications (CSA-2009), December 10 - 12, 2009
- International Conference on Parallel and Distributed Systems (ICPADS-2009), December 9 – 11, 2009
- International Conference on Network and Parallel Computing (NPC-2009), October 19-21, 2009
- International Conference on Advances in Peer-to-Peer Systems (AP2PS-202009), October 11-16, 2009
- International Conference on Embedded Software and Systems (ICESS-2009), May 25-27, 2009
- International Workshop on Service-Oriented P2P Networks (ServP2P-2009), May 18-21, 2009
- International Conference on Grid and Pervasive Computing (GPC-2009), May 4-8, 2009
- International Conference on Information Systems, Technology and Management

(ICISTM-2009), March 12-13, 2009

- High Performance Computing Asia 2009, March 2-6, 2008
- IEEE International Conference on High Performance Computing (HiPC-2008), December 17-20, 2008
- International Conference on Embedded and Ubiquitous Computing Conference (EUC-2008), December 17-20, 2008
- International Symposium on Ubiquitous Multimedia Computing (UMC-2008), October 14-16, 2008
- IEEE International Symposium on Embedded Computing (SEC-2008), October 6-8, 2008
- International Conference on Network and Parallel Computing (NPC-2008), September 18-21, 2008
- International Conference on Contemporary Computing (IC3-2008), August 7-9, 2008
- International Conference on Embedded Software and Systems (ICESS-2008), July 28-31, 2008
- International Conference on Computational Science and Engineering (CSE-2008), June 25-27, 2008
- International Conference on Communications, Circuits, and Systems (ICCCAS-2008), May 25-27, 2008
- International Conference on Grid and Pervasive Computing (GPC-2008), May 25-28, 2008
- International Conference on Advanced Information Networking and Applications (AINA-2008), March 25-28, 2008
- International Conference on Information Systems, Technology and Management (ICISTM-2008), March 6-8, 2008
- International Conference on Embedded and Ubiquitous Computing Conference (EUC-2007), December 10-13, 2007
- International Conference on Grid and Pervasive Computing (GPC-2007), May 2-4, 2007

University and Departmental Service at NTHU

- Deputy director of library, December 2003 – July 2012
- Deputy director of computer and communication research center, August 2008 – July 2015
- Ph.D. admission committee member, August 2002 – July 2015
- Master admission committee member, August 2002 – July 2015
- Faculty recruiting committee member, August 2002 – July 2009

University and Department Service at FCU

- Chairman of Department of Information Engineering and Computer Science, August 1998 – July 2001
- Network Division Director of Computer Center, January 1994 – July 1997

- Education Division Director of Computer Center, August 1993 – December 1993
 - PhD admission committee member, August 1994 – July 2002
 - Master admission committee member, August 1992 – July 2002
-

Teaching Activities

Courses taught at CUHKSZ

- CSC3050 – Computer Architecture,
- CSC3150 – Operating Systems,
- CSC4005 – Distributed and Parallel Computing,
- CSC4180 – Compiler Construction
- CSC6230 – Advanced Operating Systems,

Courses taught at NTHU

- CS135502 – Introduction to C Programming Language,
 - CS235102 – Data Structure,
 - CS242202 – Assembly Language and System Programming,
 - CS340402 – Compiler Construction,
 - CS342302 – Operating Systems.
 - CS510400 – Embedded Software for Networked SoC Systems,
 - CS510500 – Embedded Processor Design,
 - CS541000 – Virtualization Techniques,
 - CS542100 – Cloud Computing,
 - CS542200 – Parallel Programming,
 - CS542800 – Heterogeneous System Architecture.
-